BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

Z(→ℓℓ) + b/c jets in Run 3: first look at data

Summer23

2023 C data vs MC, a small sample (0.6 fb^{-1} in Z→μμ, 0.6 fb^{-1} in Z→ee), AK4 jets

29 October 2026

What is new since last time

DATA

2023 C

First look at data. Muon and EGamma, 22Sep2023 reprocessing.

CAMPAIGN

Summer23

One of four campaigns run (Summer22, Summer22EE, Summer23, Summer23BPix). Each has its own corrections and its own deck; they are not combined.

CORRECTIONS

Full set

Jet energy corrections, jet veto maps, lepton energy scale, lepton and trigger SFs, all from the official POG files. Hsin-Wei's latest code merged in.

This is a small, quick run: 20 files per MC sample and 20 data files per channel and campaign, run interactively. The aim is to see whether data and MC agree before we scale up. The selection is the same as in the MC-only study: two leptons on the Z peak and at least one AK4 jet with pT > 30 GeV. No b- or c-tag requirement yet. Since the 26 October run: electron smearing bug fixed, JME jet veto recipe, trigger SF from both electrons, newest golden JSON in every campaign, WW added for Summer22, and many more kinematic plots.

Data used

Dataset (DAS)	Files	Lumisections	brilcalc [pb^{-1}]	Factor	Used [pb^{-1}]
/EGamma0/Run2023C-22Sep2023_v1-v1/NANOAOD	20	3,756	1192	0.5	596
/Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD	20	3,526	1210	0.5	605

Summer23: Z→μμ 605 pb^{-1}, Z→ee 596 pb^{-1}

The 22Sep2023 re-reconstruction in NanoAODv12, 20 files per channel spread over the eras. Z→μμ comes from the Muon PD (IsoMu24), Z→ee from the EGamma PD (Ele30_WPTight). Muon and EGamma files cover different lumisections, so each channel has its own luminosity and the MC is scaled to it.

In 2023 each PD is split in two halves (Muon0/Muon1, EGamma0/EGamma1) that share every lumisection, each half holding about 50% of the events. We use only the first half and count half of its luminosity (factor 0.5).

Golden JSON and luminosity

GOLDEN JSON

Cert_Collisions2023_366442_370790_Golden.json

The official 2023 golden JSON, read from the CMS central area on cvmfs (cms-griddata …/DC/Collisions23/latest). This is the newest version, after the 2026 tracker re-review.

LUMINOSITY

brilcalc, normtag_PHYSICS

Run in the BRIL container on LPC, on exactly the lumisections we processed. Muon and EGamma get their own numbers.

Checked against the newest golden JSON

All 7,017 processed lumisections are golden in the newest version.

MC samples

Sample	Generator	Run3Summer23NanoAODv12-…	σ [pb]	Files
DYto2L-2Jets_MLL-50	amcatnloFXFX-pythia8	130X_mcRun3_2023_realistic_v14-v2	6346	20 / 506
DYto2L-2Jets_MLL-50_PTLL-100	amcatnloFXFX-pythia8	130X_mcRun3_2023_realistic_v15-v2	105	20 / 815
DYto2L-2Jets_MLL-50_PTLL-200	amcatnloFXFX-pythia8	130X_mcRun3_2023_realistic_v15-v2	11.24	20 / 398
DYto2L-2Jets_MLL-50_PTLL-400	amcatnloFXFX-pythia8	130X_mcRun3_2023_realistic_v15-v3	0.6433	20 / 257
DYto2L-2Jets_MLL-50_PTLL-600	amcatnloFXFX-pythia8	130X_mcRun3_2023_realistic_v15-v2	0.09941	20 / 158
TTto2L2Nu	powheg-pythia8	130X_mcRun3_2023_realistic_v14-v2	98.88	20 / 198
WW	pythia8	130X_mcRun3_2023_realistic_v14-v2	75.9	20 / 125
WZ	pythia8	130X_mcRun3_2023_realistic_v14-v2	27.6	20 / 60
ZZ	pythia8	130X_mcRun3_2023_realistic_v14-v2	12.17	20 / 11

Full DAS name: /<sample>_TuneCP5_13p6TeV_<generator>/Run3Summer23NanoAODv12-<processing>/NANOAODSIM. Files: used / available. The DY pT($\ell\ell$) samples are stitched: each binned sample keeps only its own LHE pT($\ell\ell$) slice and the inclusive one covers pT($\ell\ell$) < 100 GeV. DY events are split by the hadron flavour of the leading jet (b, c or light). $t\bar{t}$ and dibosons are not split. Cross sections as in the MC-only study.

Corrections and scale factors

Correction	Applied to	POG file	Key / tag
Jet energy corrections	data + MC	JME jet_jerc	Summer23Prompt23_V4 (data: with residuals)
Jet energy resolution	MC	JME jet_jerc, jer_smear	Summer23Prompt23_RunCv1234_JRV3
MET	data + MC	from the corrected jets	type-1 PuppiMET, jets with EM fraction < 0.9
Jet veto map	data + MC	JME jetvetomaps	Summer23Prompt23_RunC_V1
Muon momentum scale and resolution	data + MC	MUO muon_scalesmearing	Run 3 recipe
Electron energy scale / smearing	data / MC	EGM electronSS_EtDependent	Scale / SmearAndSyst
Pileup weight	MC	LUM puWeights	Collisions2023_366403_369802_eraBC_GoldenJson
Muon ID and isolation SF	MC	MUO muon_Z	NUM_TightID_DEN_TrackerMuons, NUM_TightPFIso_DEN_TightID
Muon trigger SF (leading muon)	MC	MUO muon_Z	NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
Electron ID and reco SF	MC	EGM electron	Electron-ID-SF: Tight and Reco (2023PromptC)
Electron trigger SF (either electron)	MC	EGM electronHlt	HLT_SF_Ele30_TightID
DY Z pT weight	DY MC	ours, from Z→μμ data	data/ZbZc/zpt_weights_run3.json
b- and c-tag SF	MC	BTV btagging, ctagging	not applied yet (no tag cut)

Every POG file is the official one on cvmfs: /cvmfs/cms-griddata.cern.ch/cat/metadata/<POG>/Run3-23CSep23-Summer23-NanoAODv12/latest/. The framework uses them for all BTV workflows. The Z pT weight is ours (DY MC vs Z→μμ data, before the MET and jet cuts).

POG recommendations vs what we do (1/2)

Data and jets

Topic	POG recommendation	What we do	Status
Golden JSON (DC)	Use the latest golden JSON	Latest version from cvmfs	**Same**
Luminosity (LUM)	brilcalc with normtag_PHYSICS	Same, on the lumisections we processed	**Same**
Pileup (LUM)	Official pileup weights	Same file	**Same**
Jet energy corrections (JME)	Summer23Prompt23_V4, resolution Summer23Prompt23_RunCv1234_JRV3	Same tags	**Same**
Jet ID (JME)	Tight lepton veto ID; the NanoAODv12 jetId branch needs the JME fix	Fix applied	**Same**
Jet veto map (JME)	Veto the event if a jet with pT > 15 GeV, tight ID, EM fraction < 0.9 and away from muons falls in the map	As recommended (since 28 Oct; before, any jet in the map vetoed the event and about 6% of Z→μμ events were lost instead of 1–2%)	**Same**
Rare high-pT jets (JME)	Reject jets with rawFactor > 0.9 (known JEC issue)	Not applied. About 1 event in a million, mostly forward, outside $\|\eta\| < 2.5$	**Not yet**

Read on 26 October 2026: BTV scale factor page, MUO usage recommendations, JME jet energy scale page, EGamma Run 3 TWiki (last edited 2023, older than the cvmfs files we use). Jet veto recipe from the JME jet veto map recommendation. The 6% and 1–2% are quick estimates on one chunk of the 26 October run.

POG recommendations vs what we do (2/2)

Leptons and tagging

Topic	POG recommendation	What we do	Status
Muon ID and isolation (MUO)	Tight ID, PF isolation < 0.15 (Tight), SFs from muon_Z	As recommended	**Same**
Muon trigger SF (MUO)	IsoMu24 SF per muon (Tight ID + Tight iso)	Leading muon only (pT > 35 GeV, on the plateau). Effect of using both muons not checked yet	**Simplified**
Electron ID and reco (EGM)	Cut-based Tight ID, SFs for the 22Sep2023 re-reco	As recommended	**Same**
Electron trigger SF (EGM)	HLT_SF_Ele30_TightID per electron	Both electrons, from the EGM data and MC efficiencies. Z→ee MC about 1% higher than with the leading electron alone	**Same**
Lepton energy scale (EGM, MUO)	Scale in data, smearing in MC	Same recipe. Fixed a framework bug (one smearing number per event): MC Z→ee peak was 5–10% too wide	**Same**
b/c tagging (BTV)	AK4 PUPPI, pT > 20, |η| < 2.5, tight ID, hadronFlavour. PNet b: fixed-WP and shape SFs; PNet c: fixed-WP SFs only	pT > 30, |η| < 2.5, tight ID, hadronFlavour. No tag SFs yet (no tag cut)	**Not yet**
L1 prefiring	No prefiring weight in Run 3	None	**Same**

c-tagging has no shape SFs for 2022/2023, only fixed working points. A template fit in the CvL–CvB plane needs a plan for that. UParT SFs exist only for NanoAODv15.

What Hsin-Wei's update brought

We merged her latest coffea_machine branch (5 commits, up to 1 October) into ours on 26 October, before this run.

FIXED FOR EVERYONE

Electron SFs with two electrons

The shared electron SF code used to keep only the last electron's SF. Her fix multiplies the SFs of all electrons. We had worked around it by calling it once per electron, so our numbers do not change, but the common code is now right.

SAFER

Empty lepton slots

The SF code now fills empty lepton slots before calling the POG files. The 2023 reco SFs also need the electron φ, and that path is now protected too.

TO REUSE

Matched electron trigger SF

Her Run 2 electron trigger SF takes the electron that fired the trigger, matched to the HLT object. That is the right way to do it, and we plan to bring it to Run 3 instead of using the leading electron.

Bottom line

Her Run 2 electron SF files and settings do not touch the Run 3 inputs, and both branches now share one correction code. Where files clashed we kept our validated Run 2 selection. One thing to tell her: in the muon SF code the trigger-SF pT clip gets overwritten (we do not use that path).

Cutflow: data vs MC

	leptons	OS	Zmass	MET	jet
Z→ee data	181,023	179,427	171,117	163,377	30,454
MC	180,645	179,495	171,888	160,804	30,861
data / MC	1.00	1.00	1.00	1.02	0.99
Z→μμ data	348,010	347,844	329,727	313,818	56,592
MC	351,194	351,146	337,274	316,457	58,895
data / MC	0.99	0.99	0.98	0.99	0.96

Events that fired the trigger and then passed each step. The table starts after the trigger: the triggered data also hold many single-lepton events (mostly W→ℓν) that we do not simulate, so data/MC only means something once two leptons are required. MC here has no SFs or pileup weight yet (generator weight × σ × L / Σw only); data/MC with all weights is on the next page.

Yields after the full selection

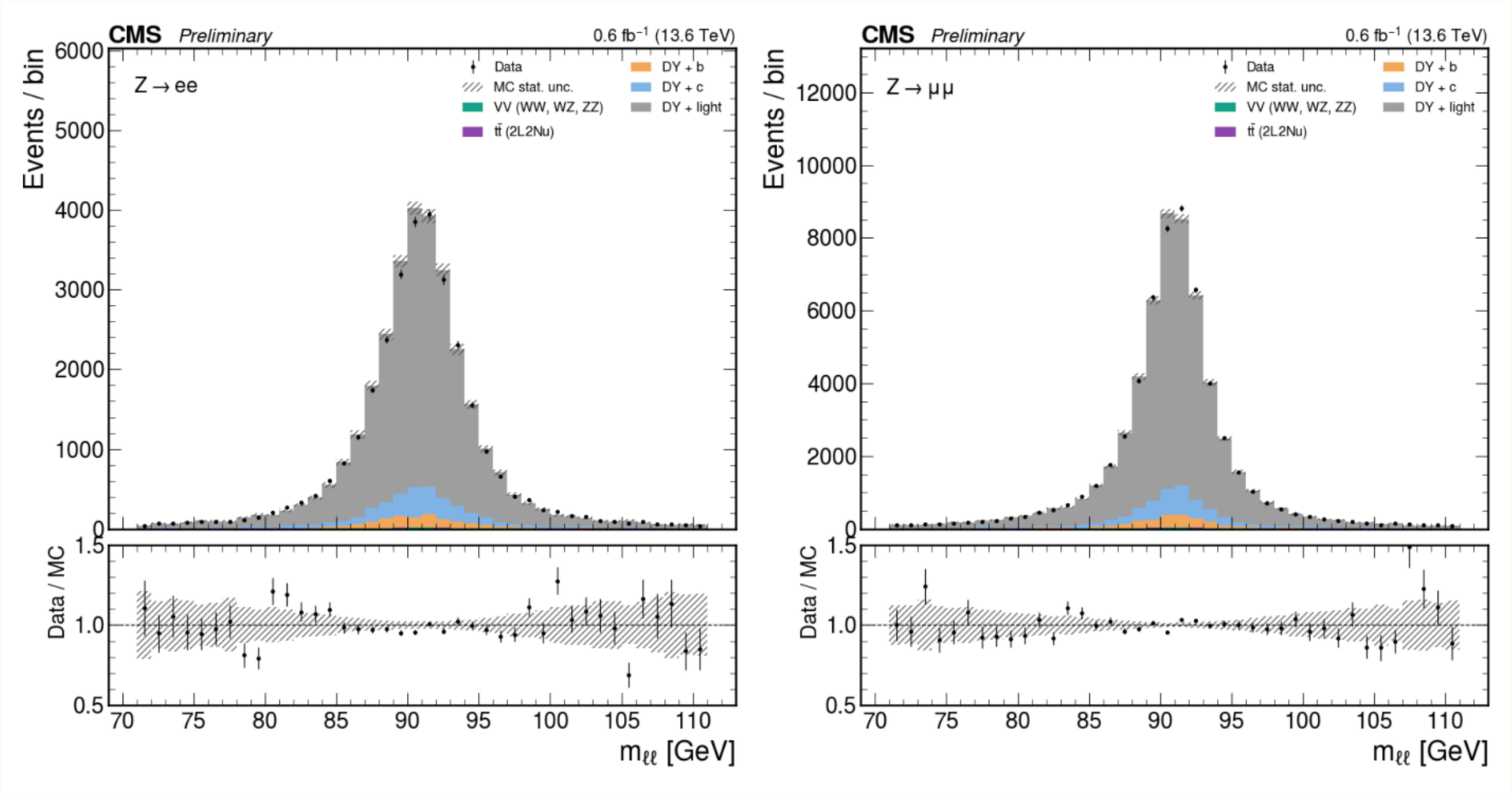
	DY + light	DY + c	DY + b	tt̄	VV	MC	Data	Data/MC
Z→ee	26,836	2,651	1,200	119	103	30,909	30,454	0.985
Z→μμ	49,211	4,770	2,260	216	181	56,638	56,592	0.999

Z→ee: DY + b is 3.9% and DY + c is 8.6% of DY (c/b = 2.21); tt̄ + VV is 0.7% of the MC.

Z→μμ: DY + b is 4.0% and DY + c is 8.5% of DY (c/b = 2.11); tt̄ + VV is 0.7% of the MC.

Dilepton mass

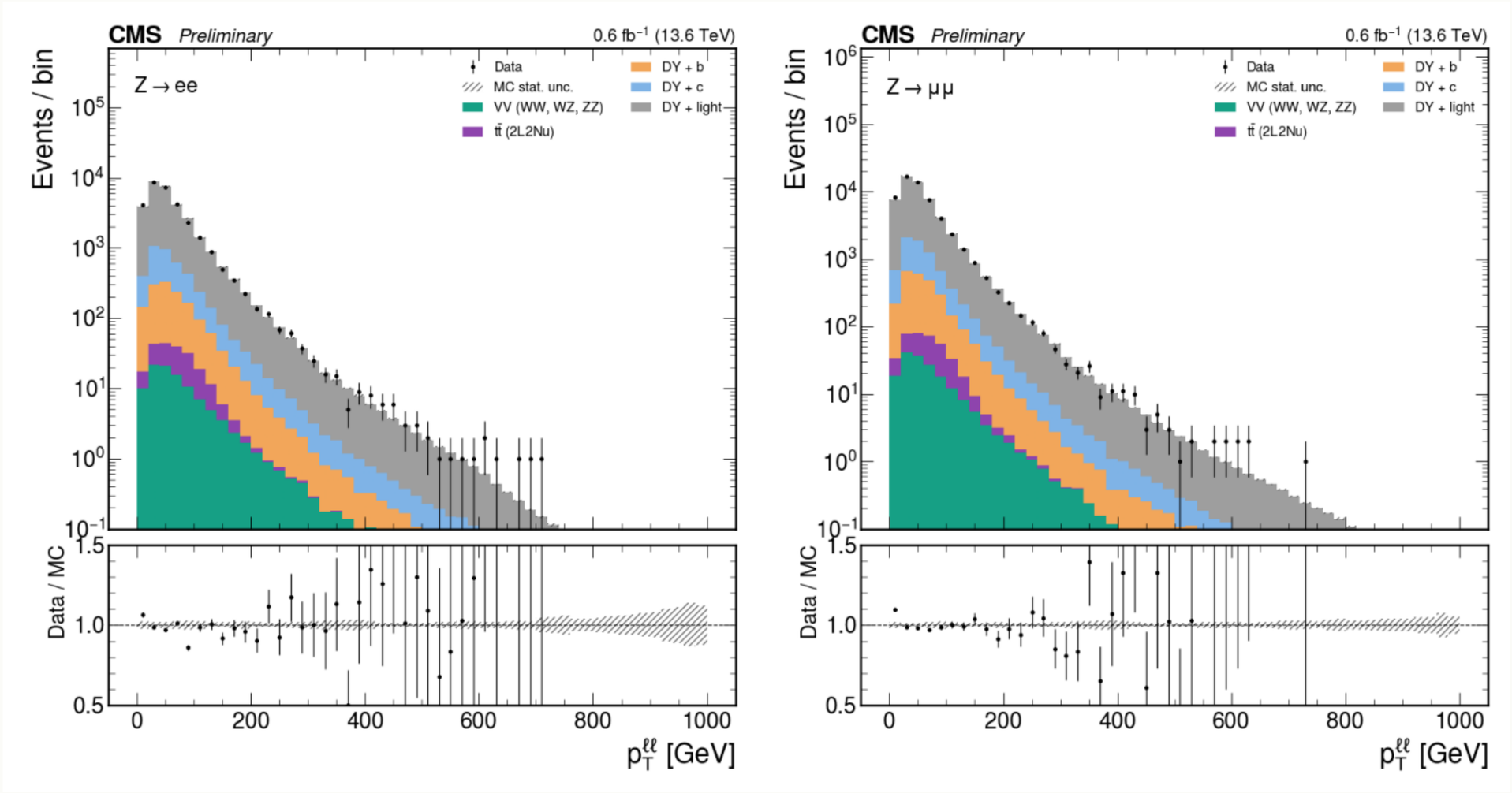
The Z peak in data and MC. Linear y-axis.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 12

Dilepton pT

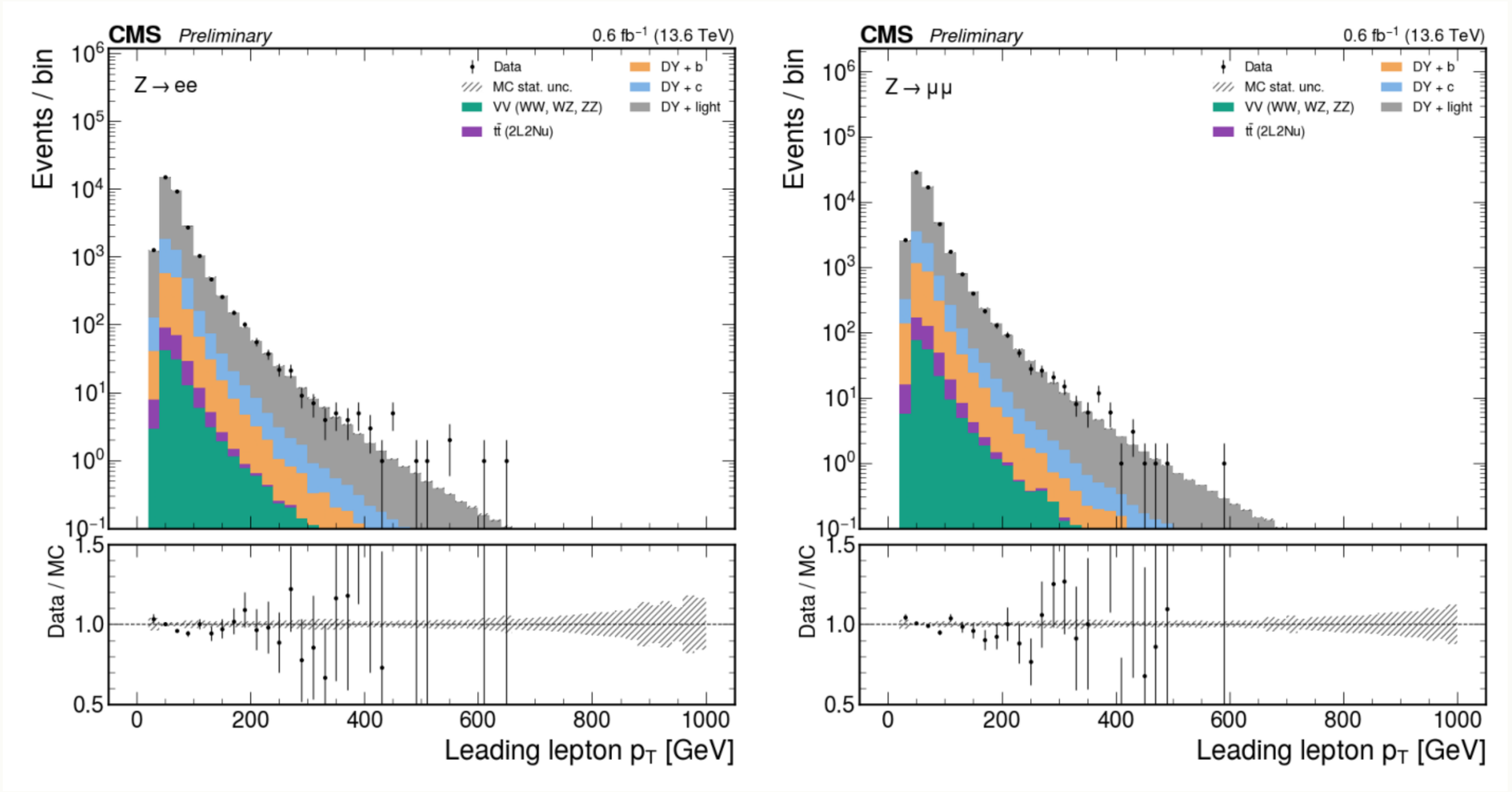
pT of the Z candidate. No cut on it.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 13

Leading-lepton pT

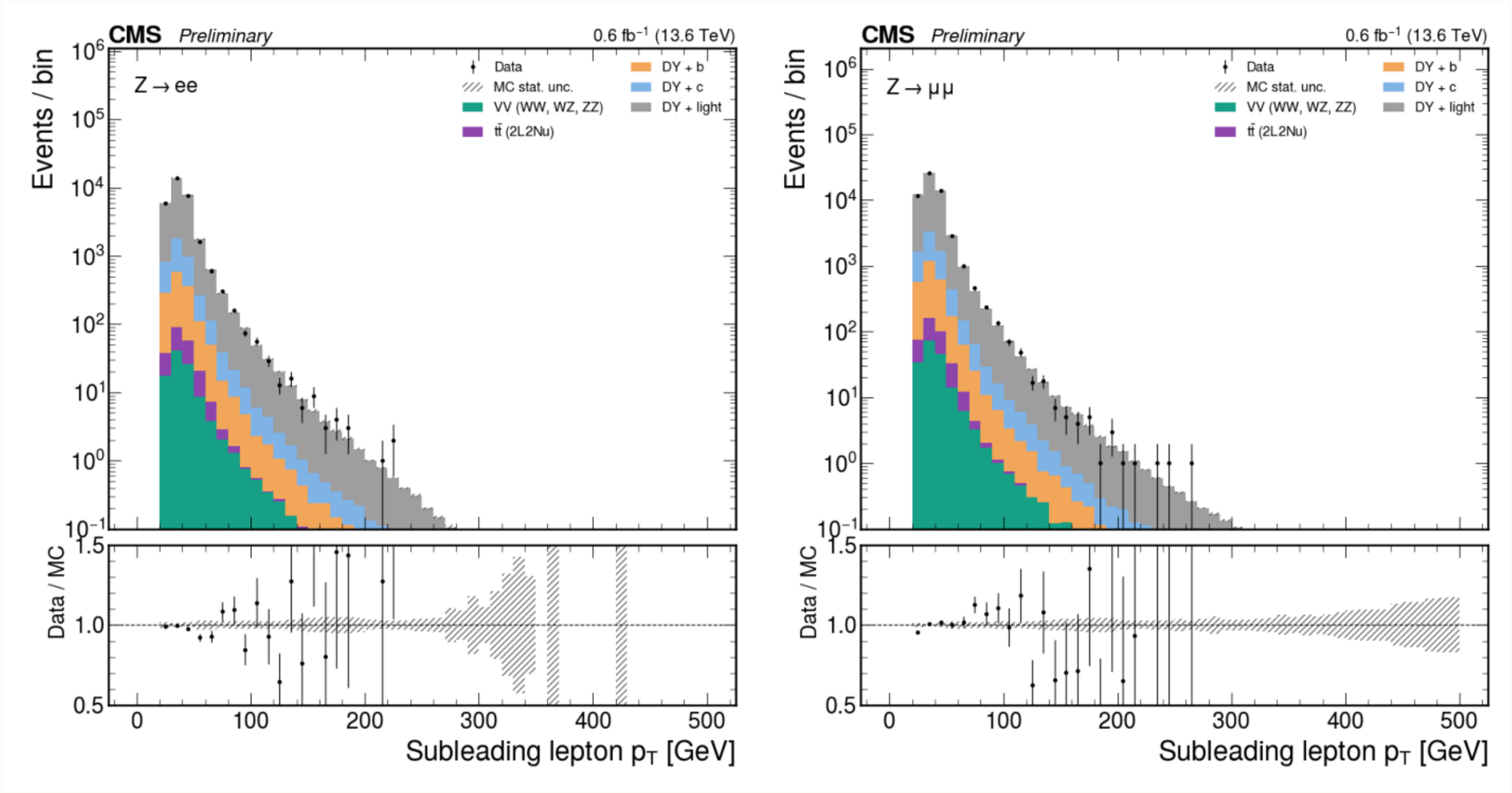
Leading lepton, pT > 35 GeV required.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 14

Subleading-lepton pT

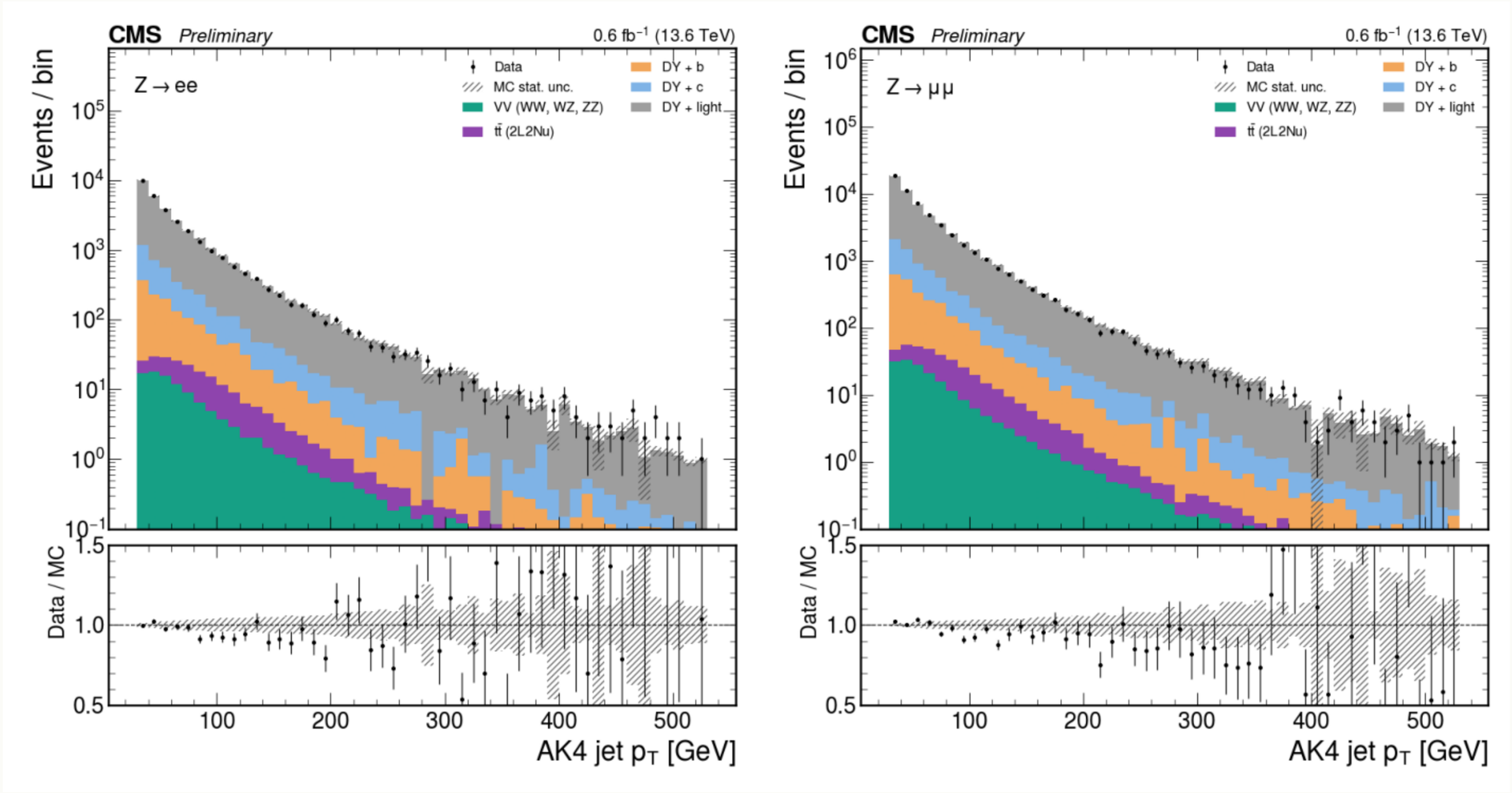
Subleading lepton, pT > 25 GeV required.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 15

Leading AK4 jet pT

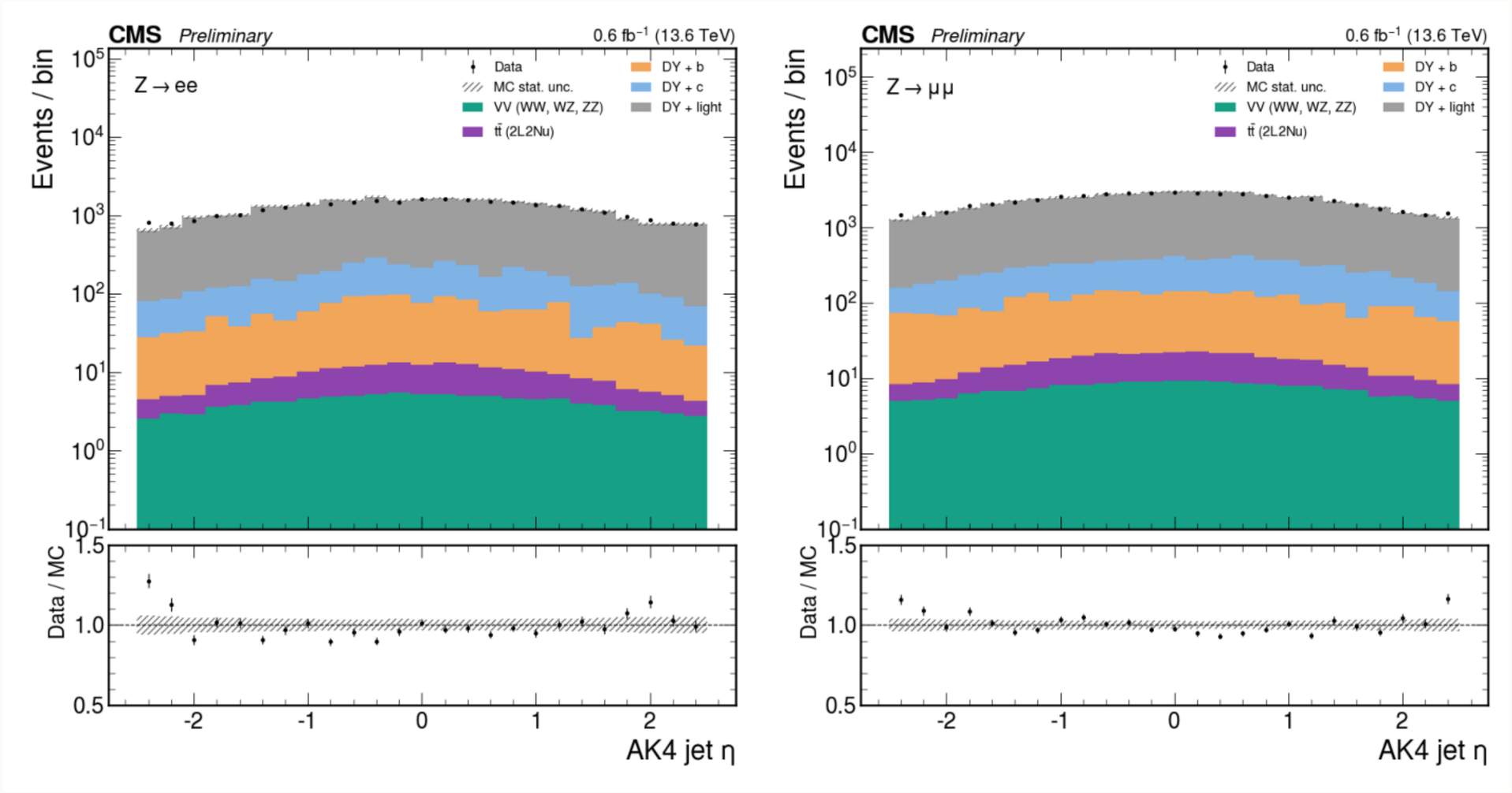
Leading lepton-cleaned AK4 jet, pT > 30 GeV required.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 16

Leading AK4 jet η

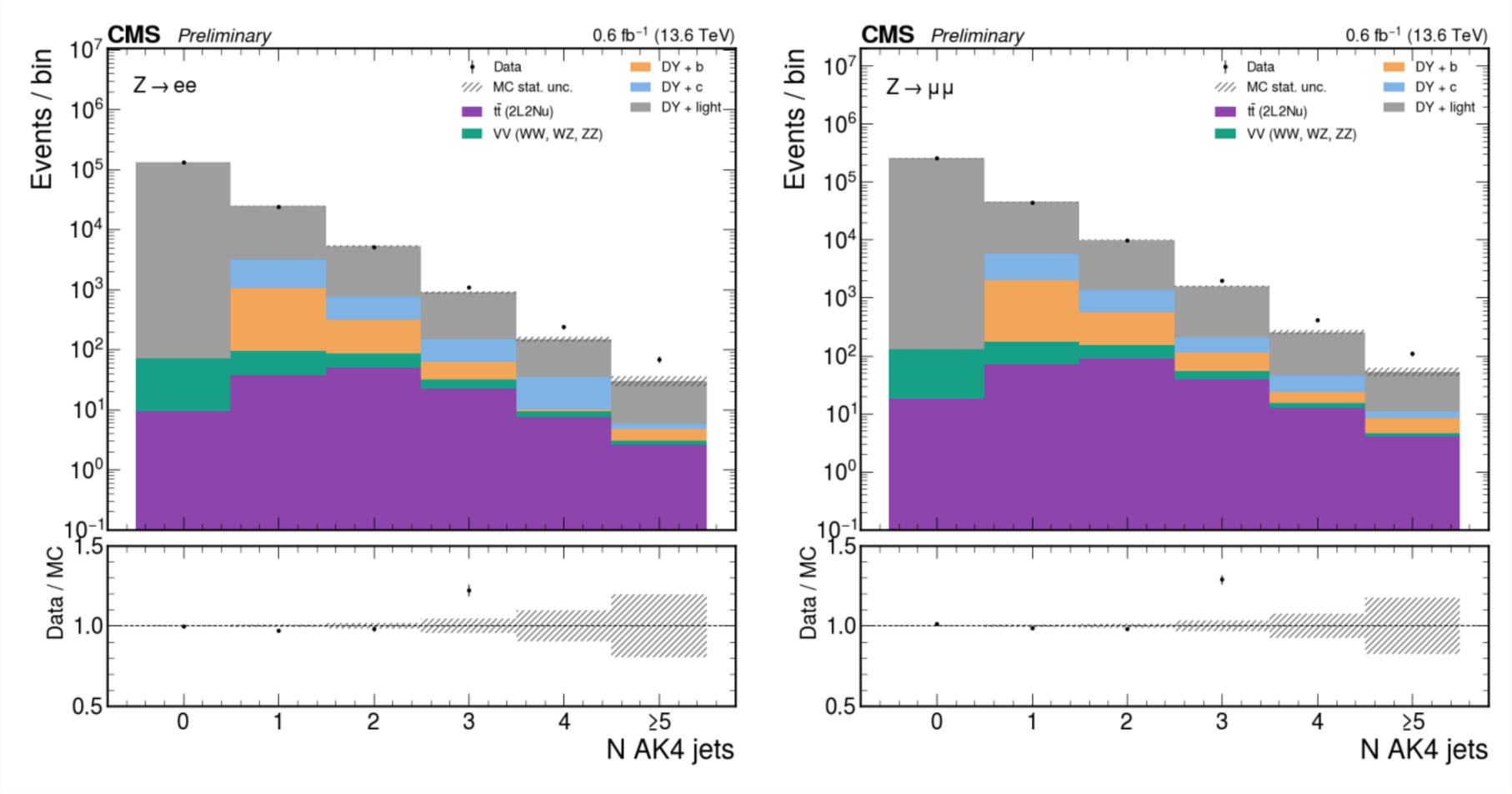
Pseudorapidity of the leading jet, |η| < 2.5.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 17

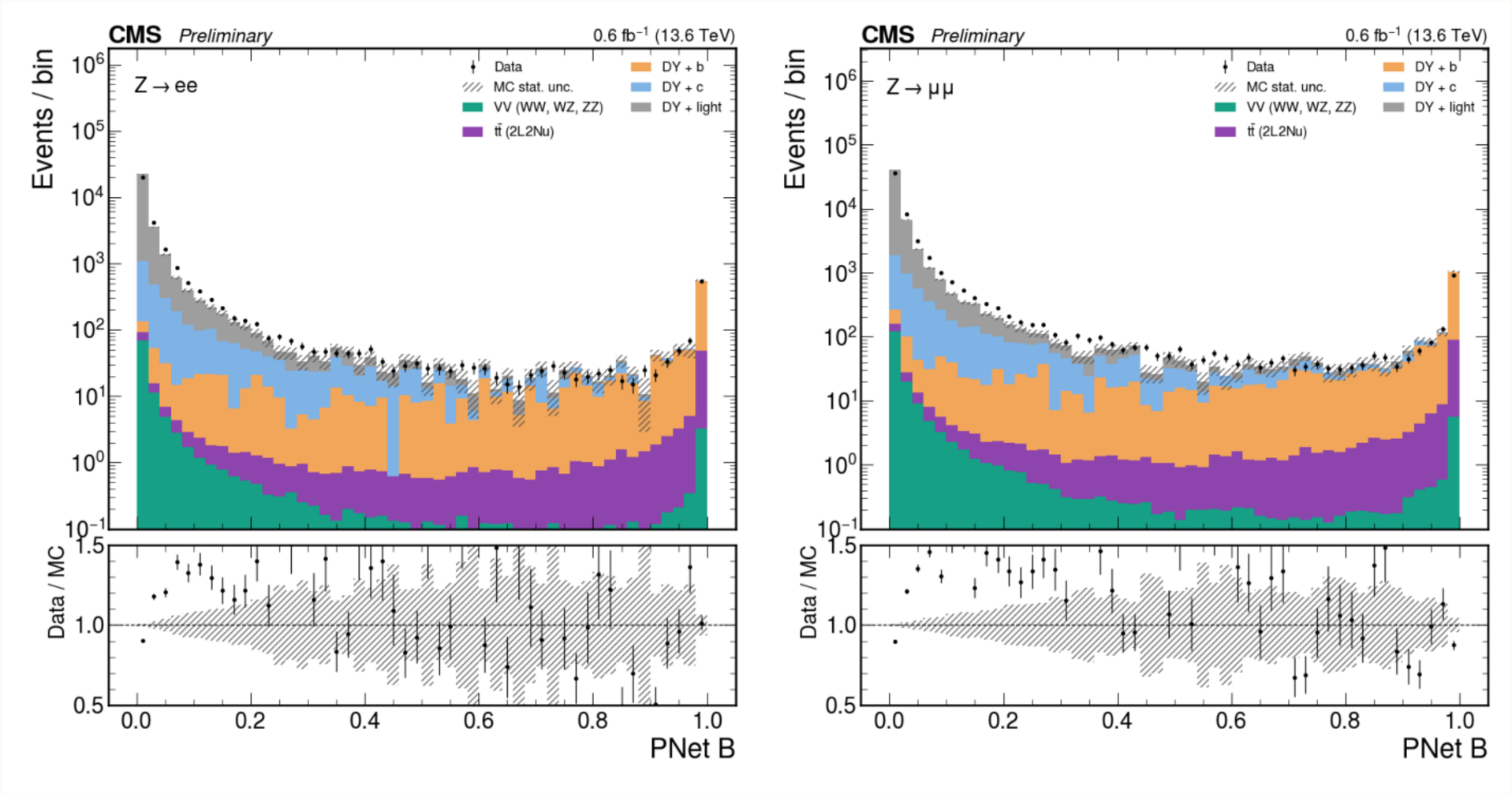
Number of AK4 jets

Jets with pT > 30 GeV and |η| < 2.5. The last bin holds 5 or more jets. Filled before the at-least-one-jet cut, so the 0 bin holds the Z events without a jet.



PNet b-tag score

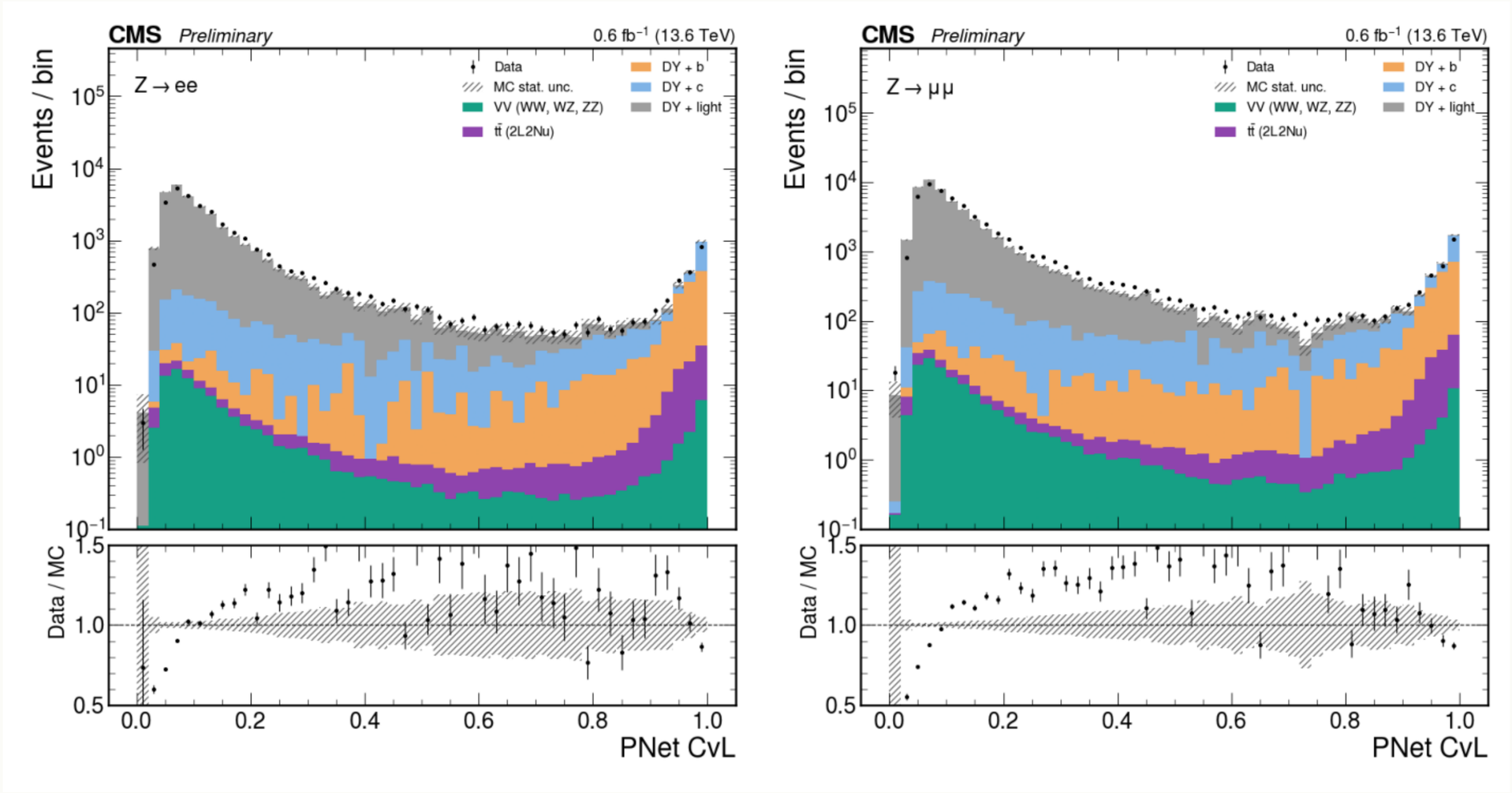
Leading jet. b jets sit near 1, light jets near 0.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 19

PNet charm vs light

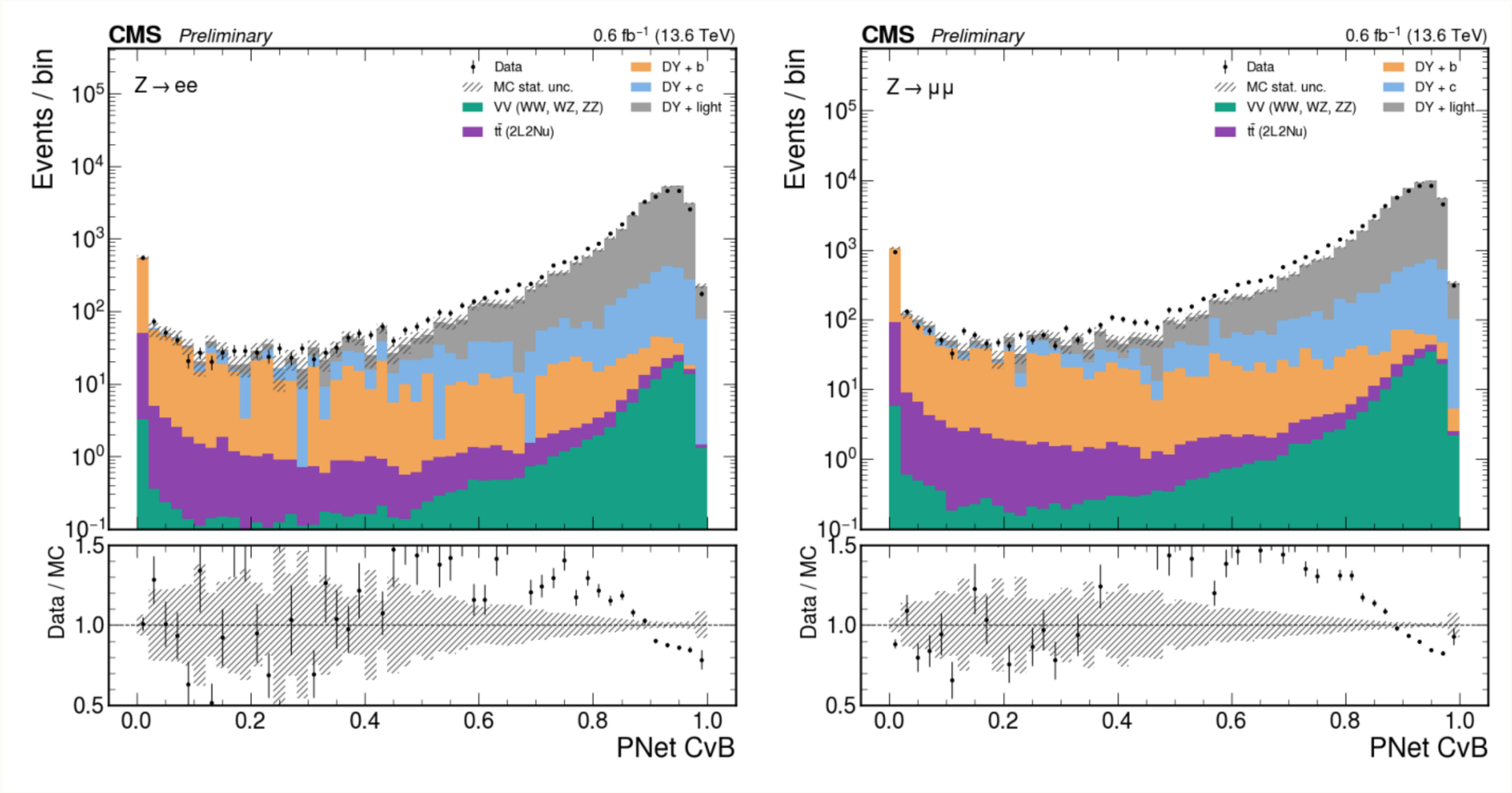
Leading jet.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 20

PNet charm vs b

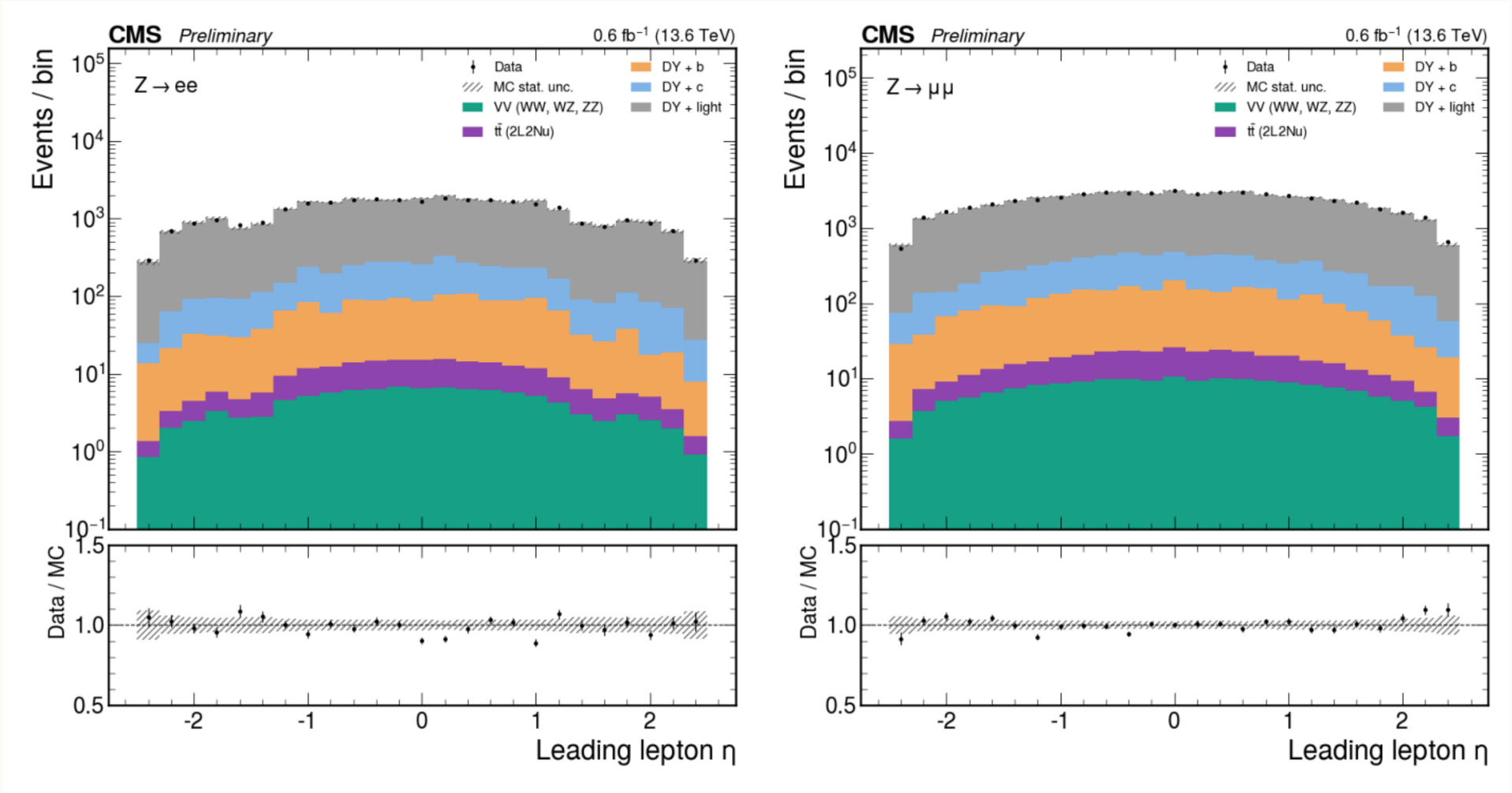
Leading jet.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 21

Leading-lepton η

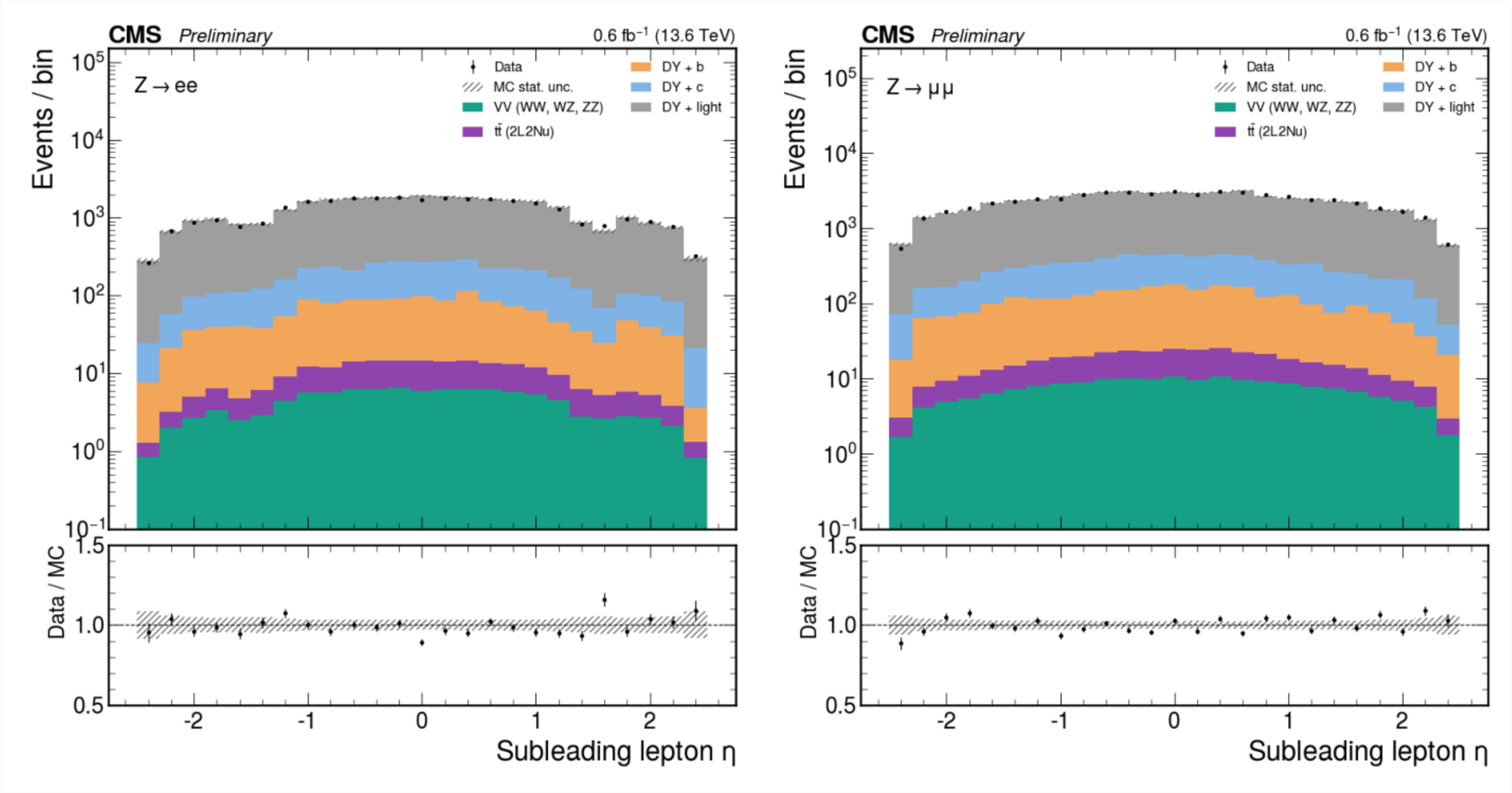
Pseudorapidity of the leading lepton.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 22

Subleading-lepton η

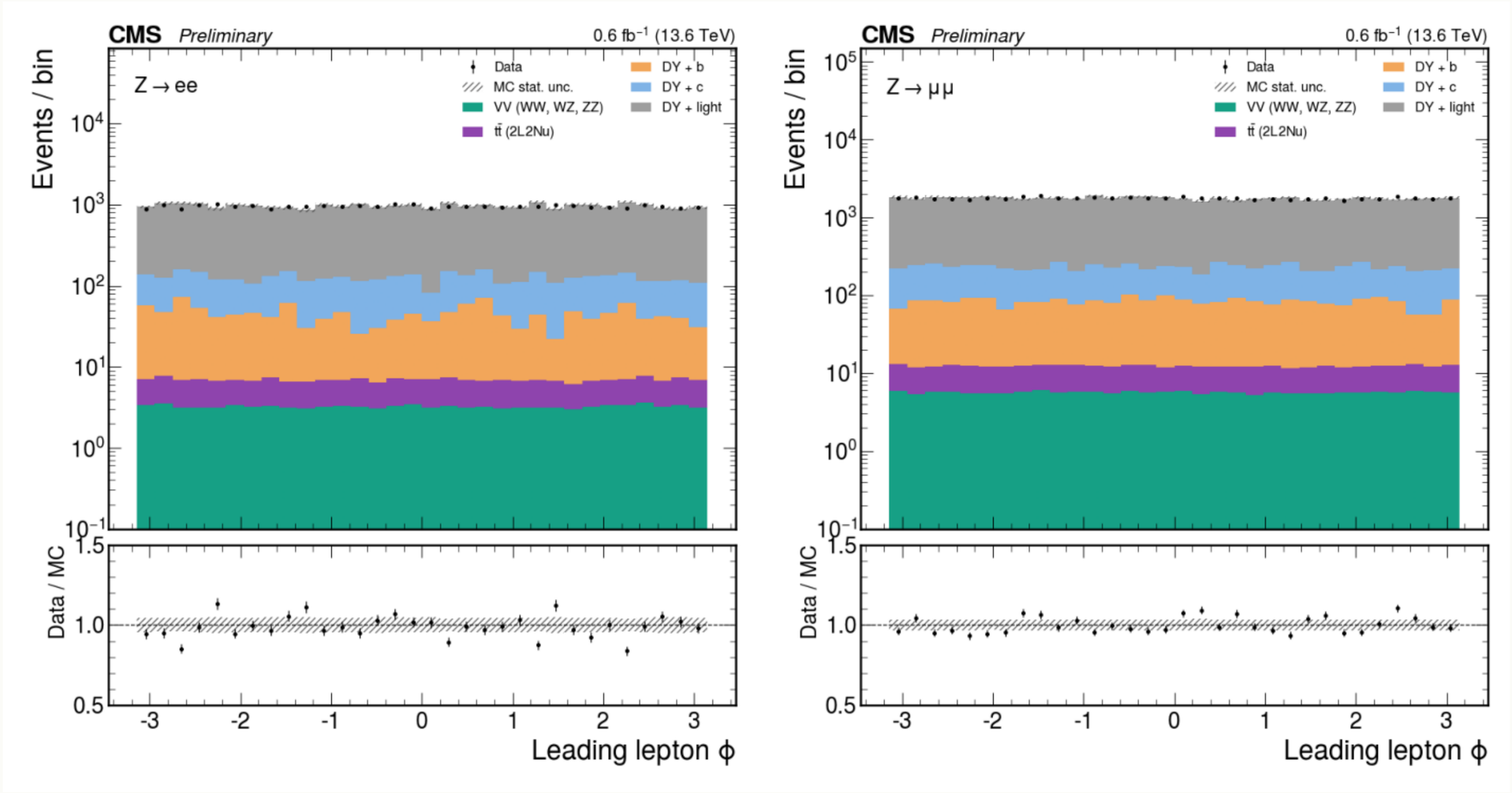
Pseudorapidity of the subleading lepton.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 23

Leading-lepton φ

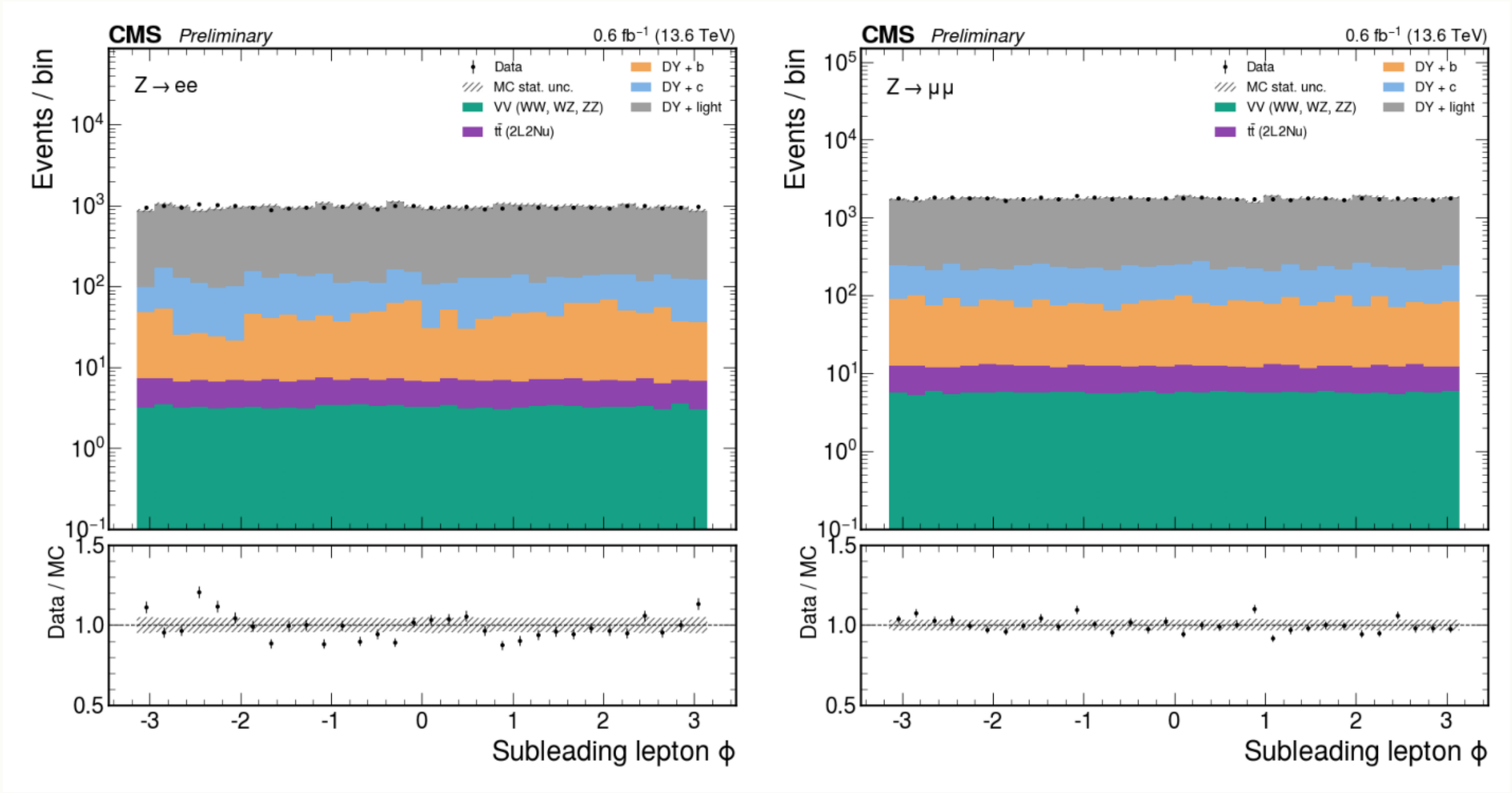
Flat in a perfect detector. Dips point to detector holes.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 24

Subleading-lepton φ

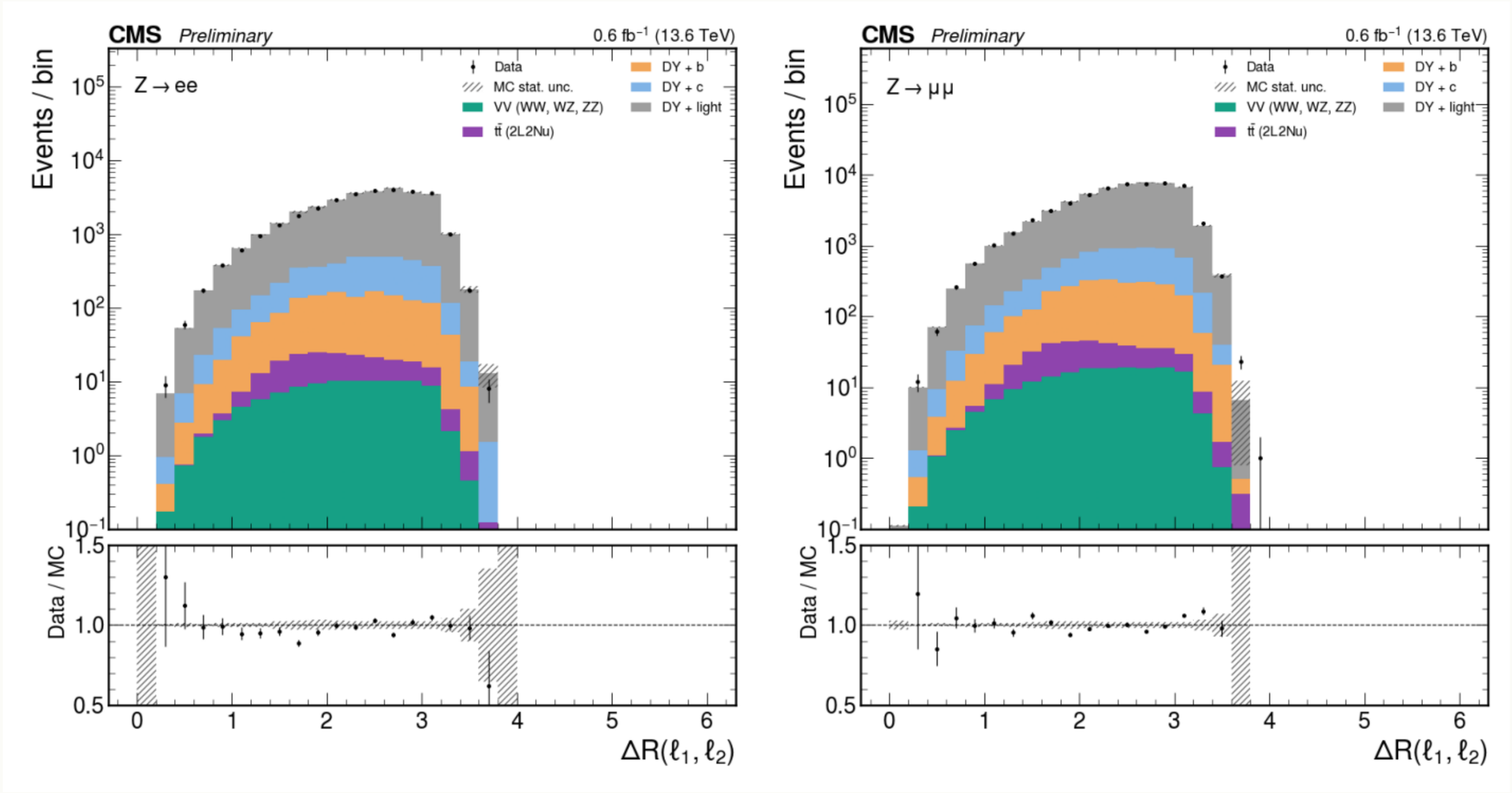
Flat in a perfect detector. Dips point to detector holes.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 25

ΔR between the two leptons

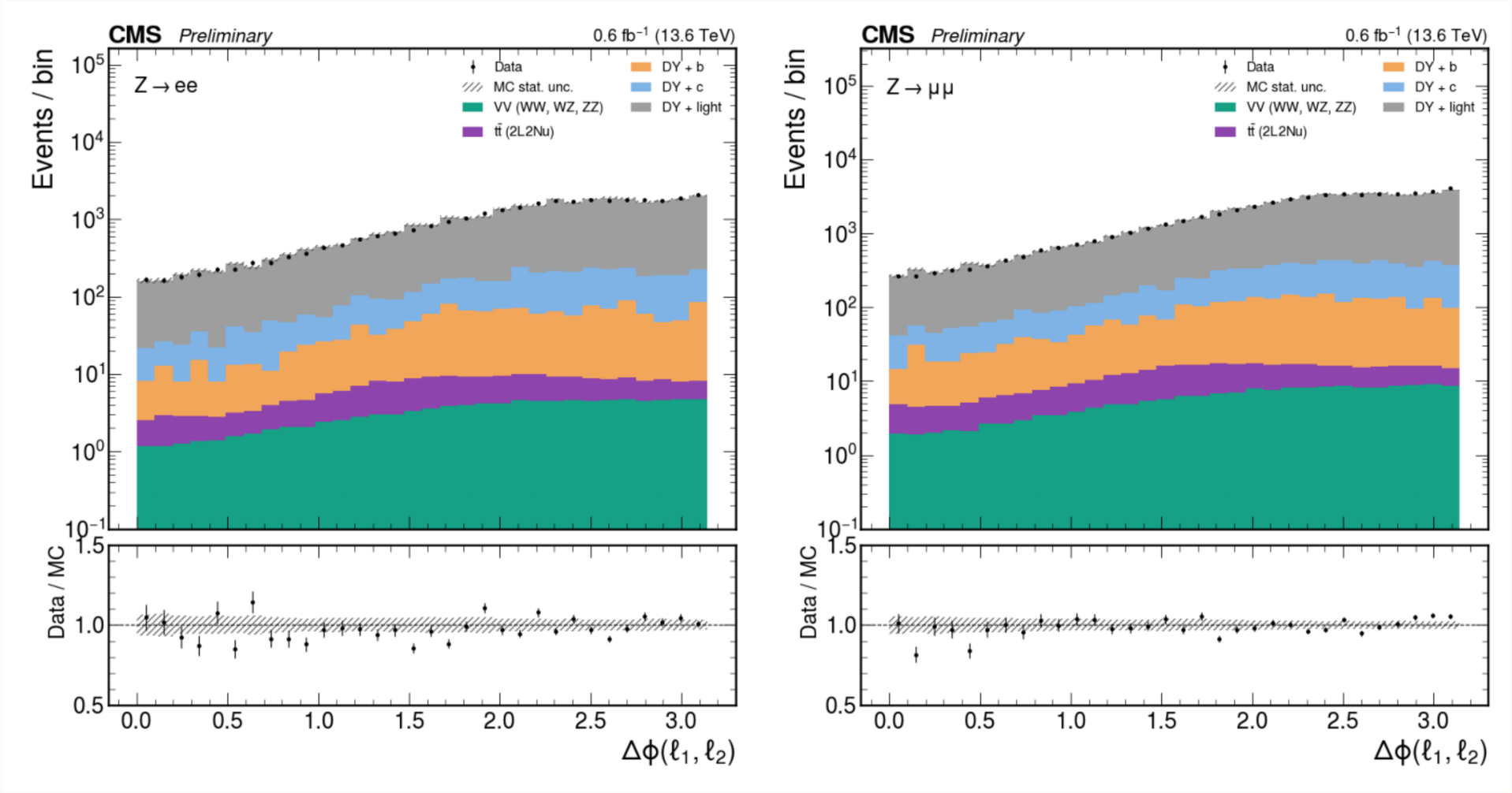
Small ΔR means a boosted Z.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 26

Δφ between the two leptons

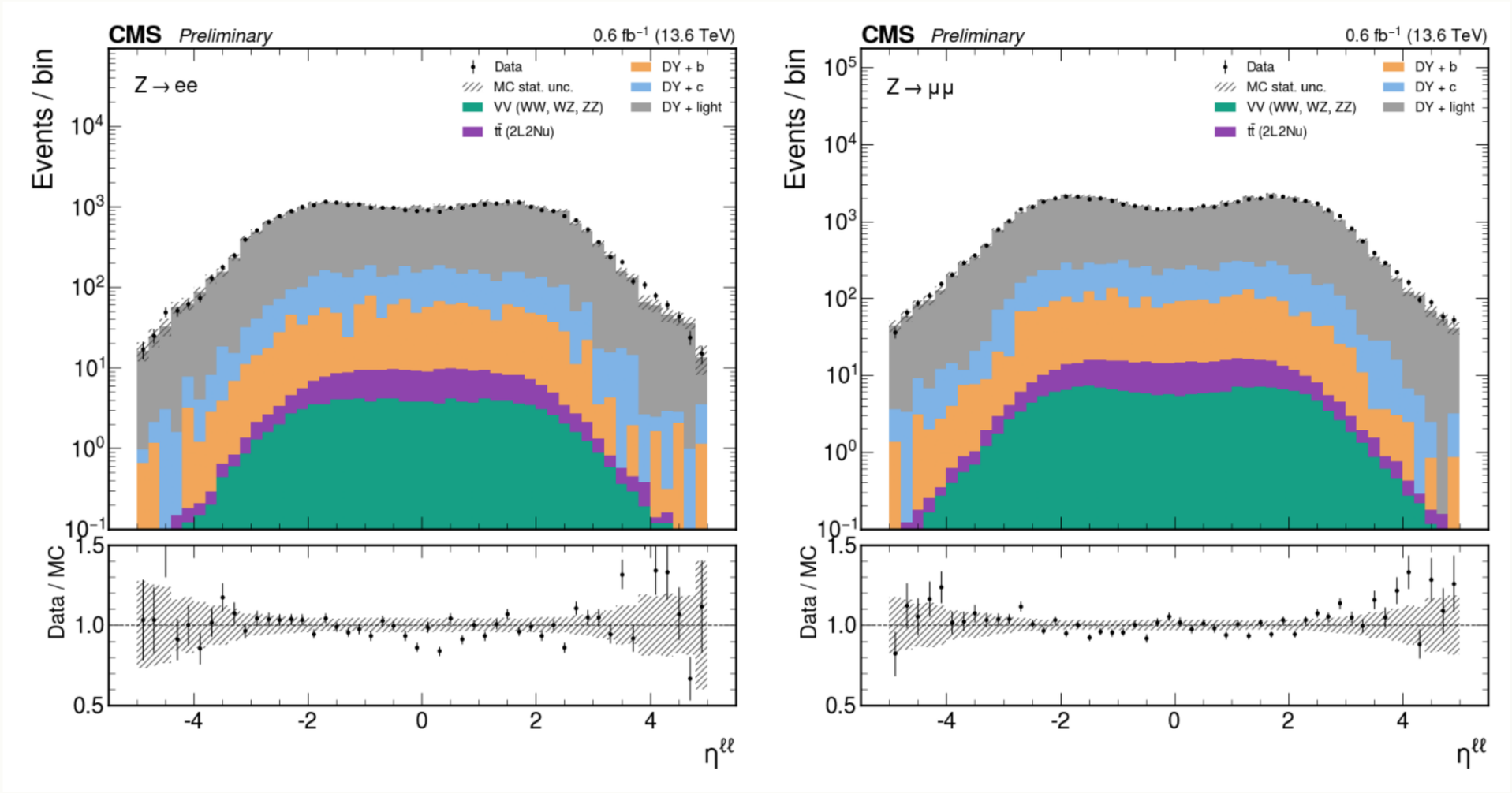
Back-to-back leptons sit at π.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 27

Dilepton η

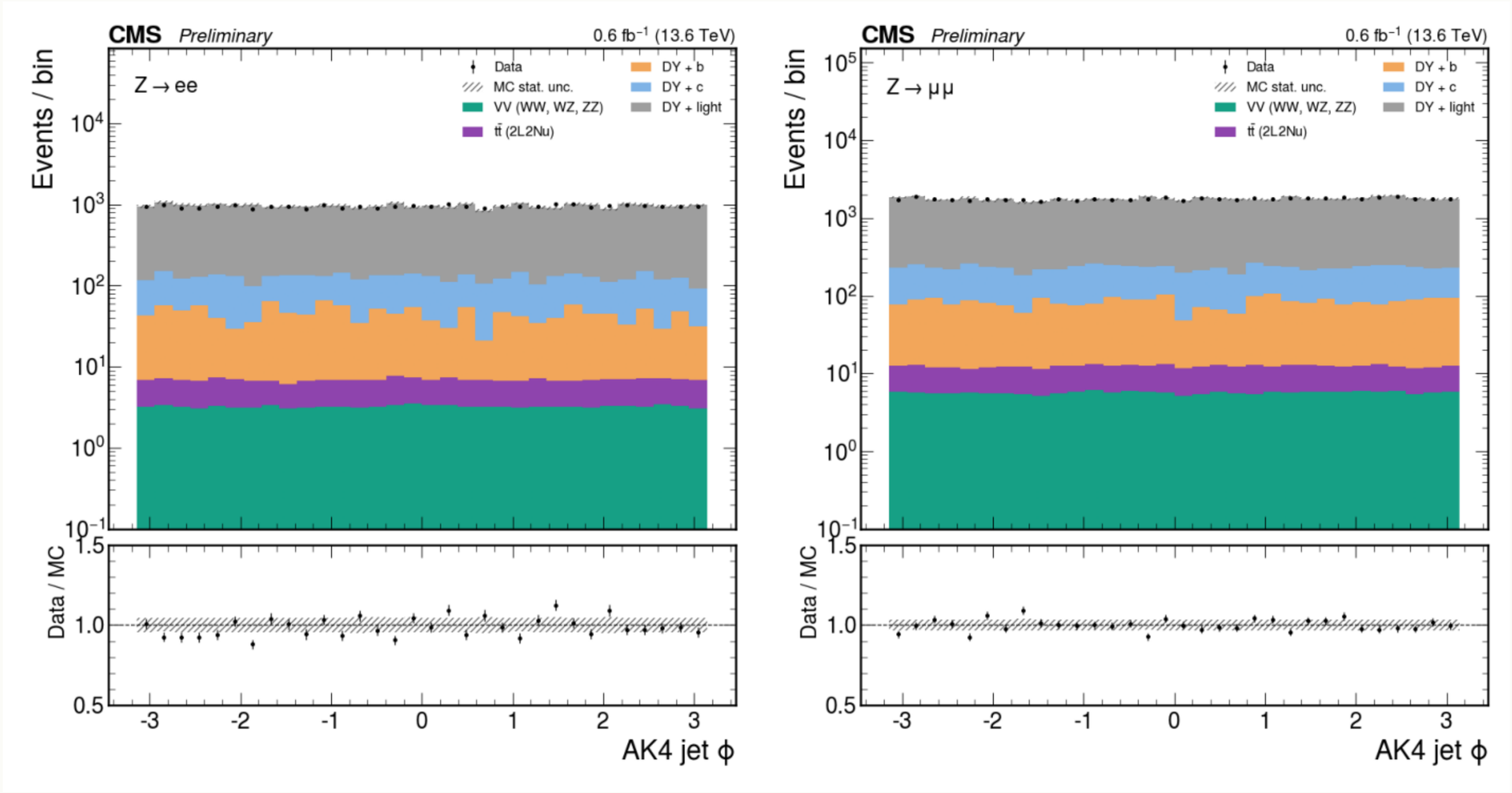
Pseudorapidity of the Z candidate.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 28

Leading AK4 jet φ

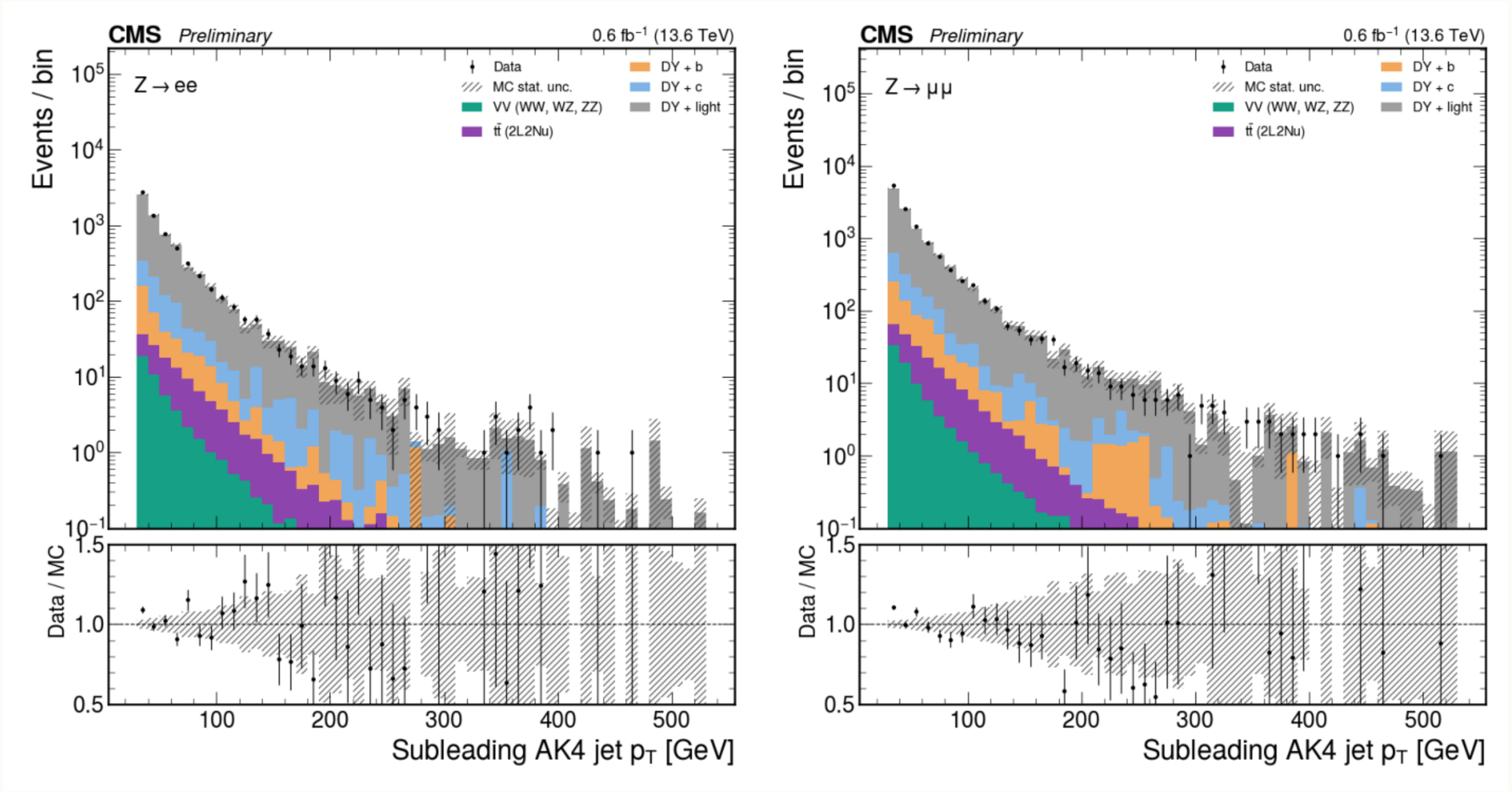
Flat in a perfect detector. The jet veto map removes some regions.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 29

Subleading AK4 jet pT

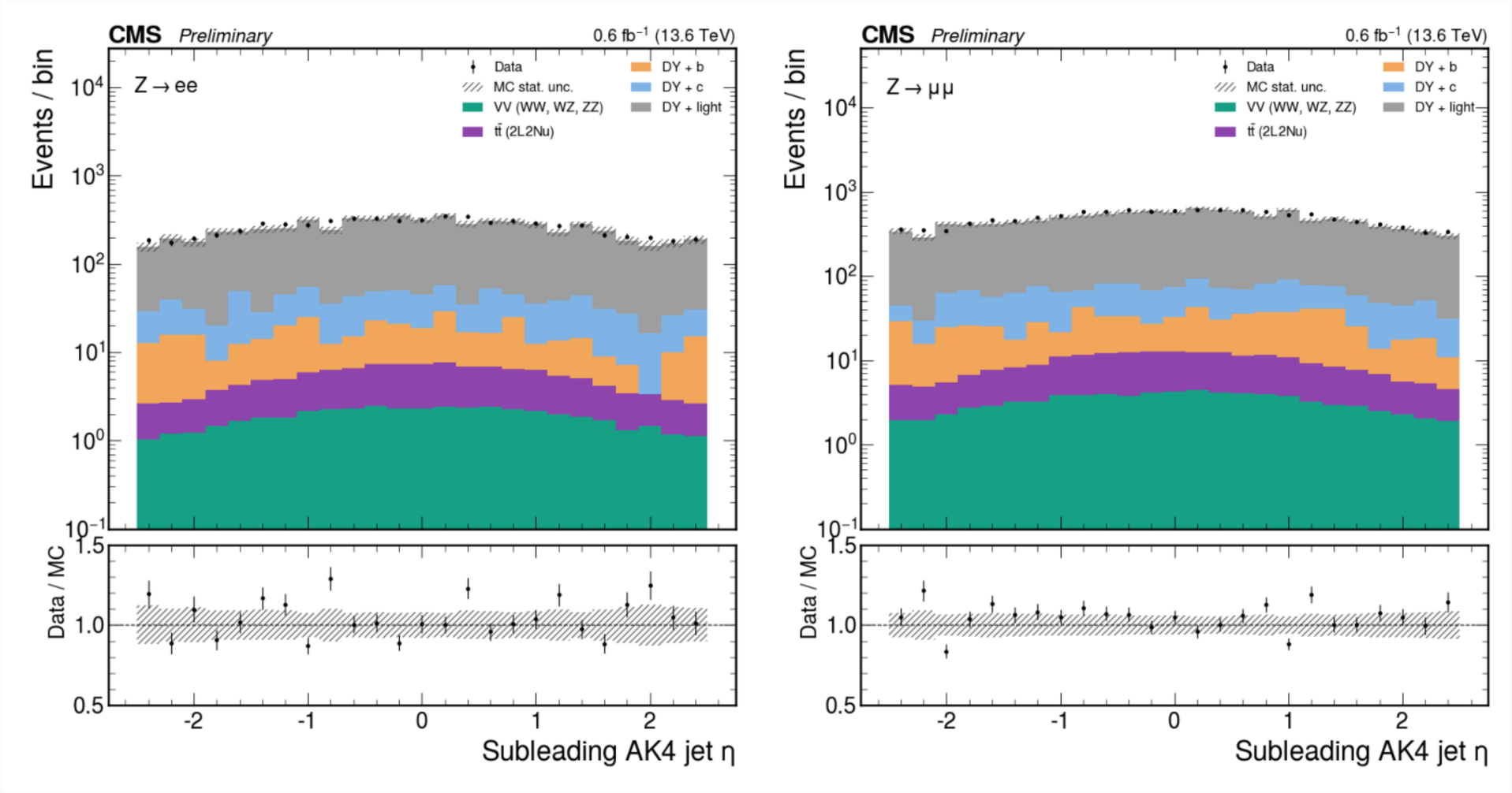
Events with at least two jets.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 30

Subleading AK4 jet η

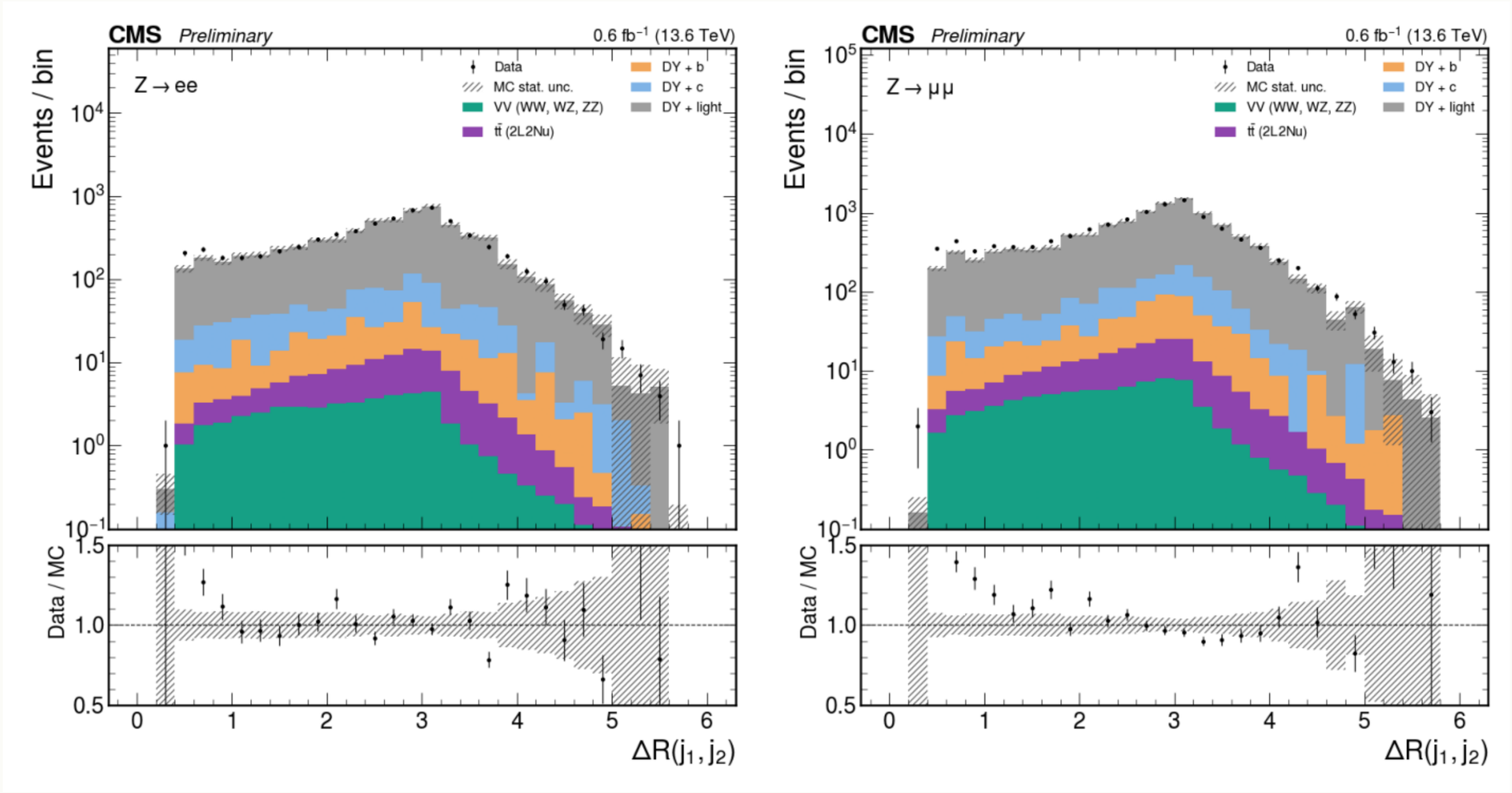
Events with at least two jets.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 31

ΔR between the two leading jets

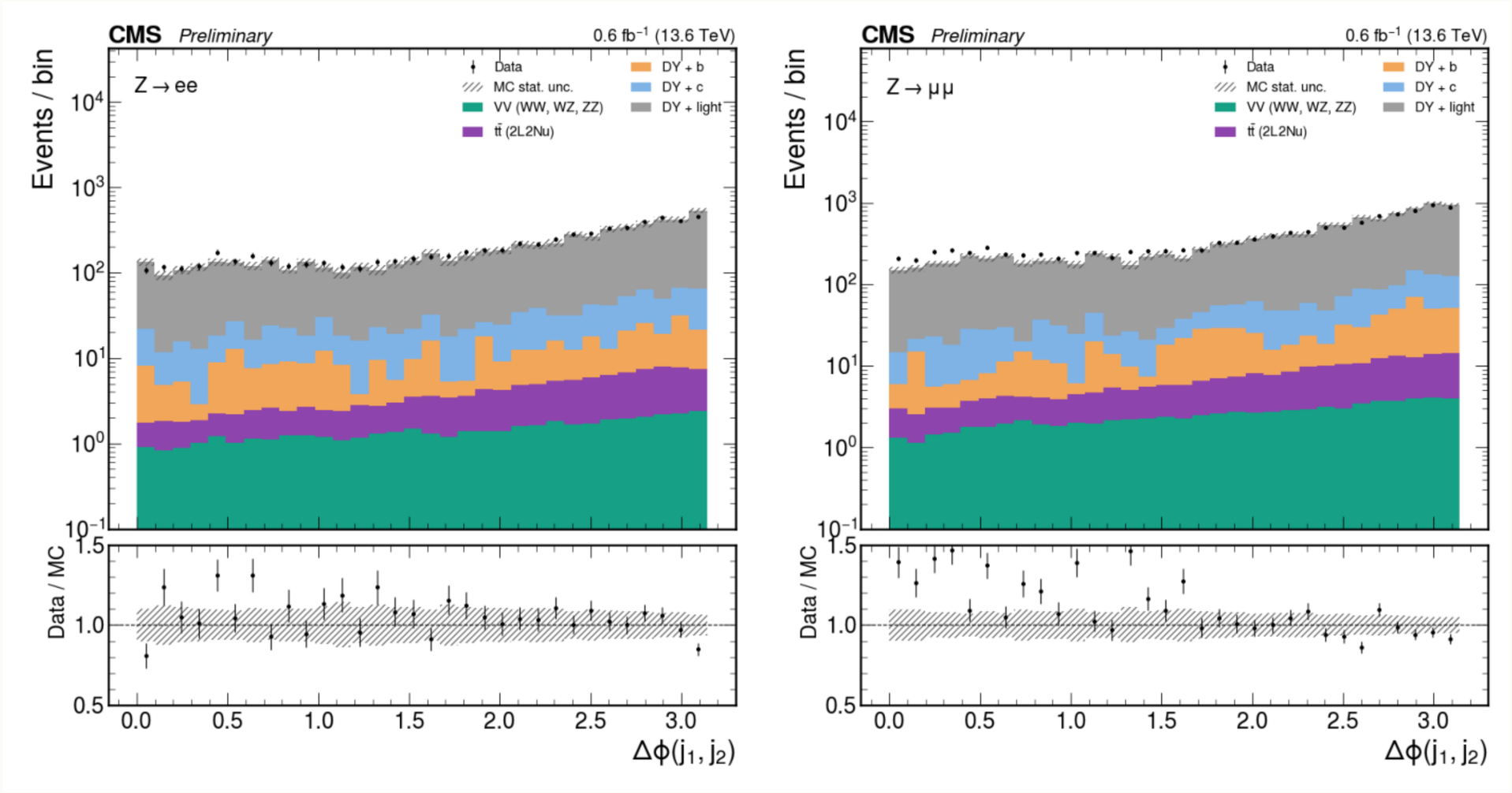
Events with at least two jets.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 32

Δφ between the two leading jets

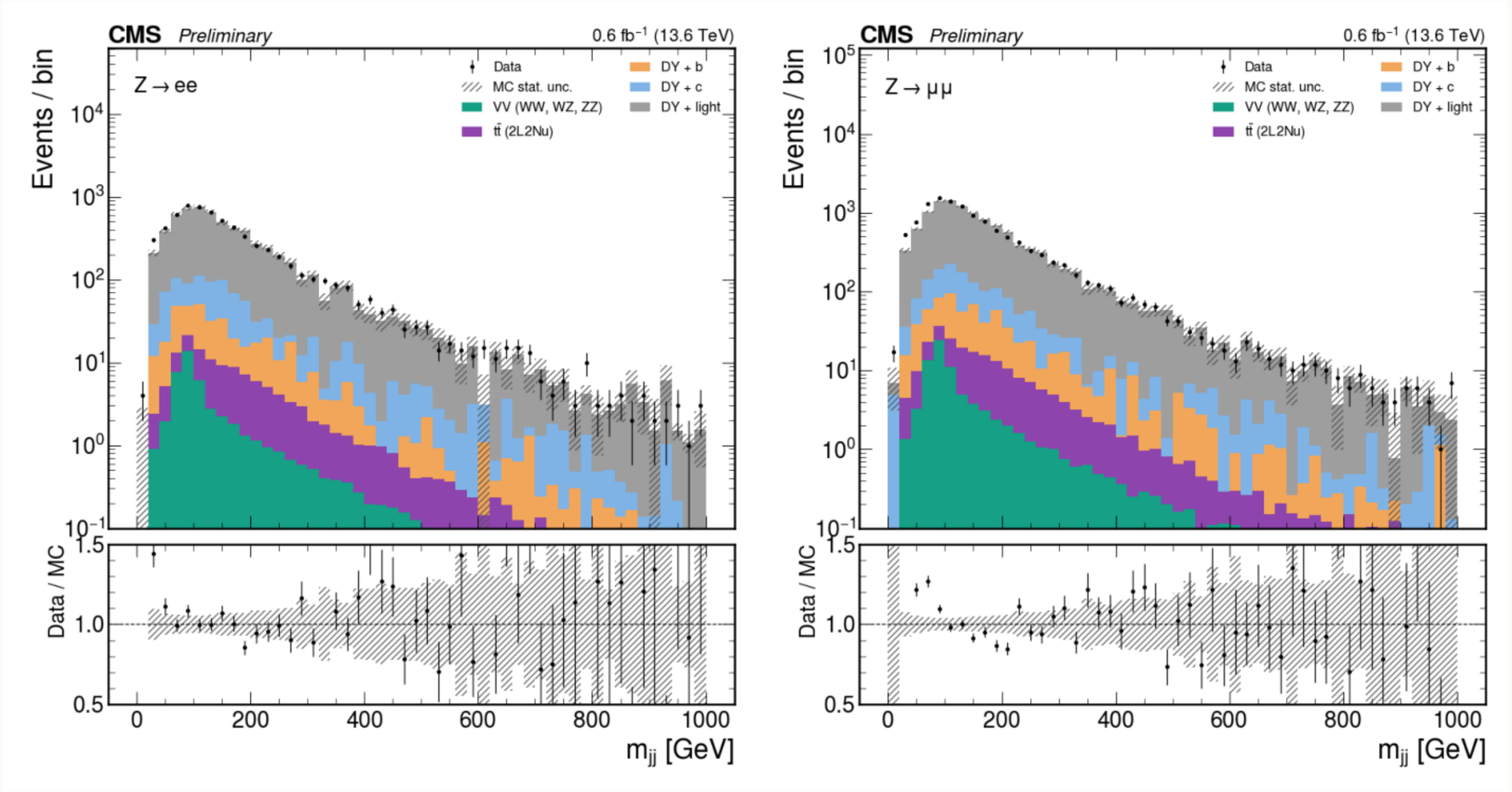
Events with at least two jets.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 33

Mass of the two leading jets

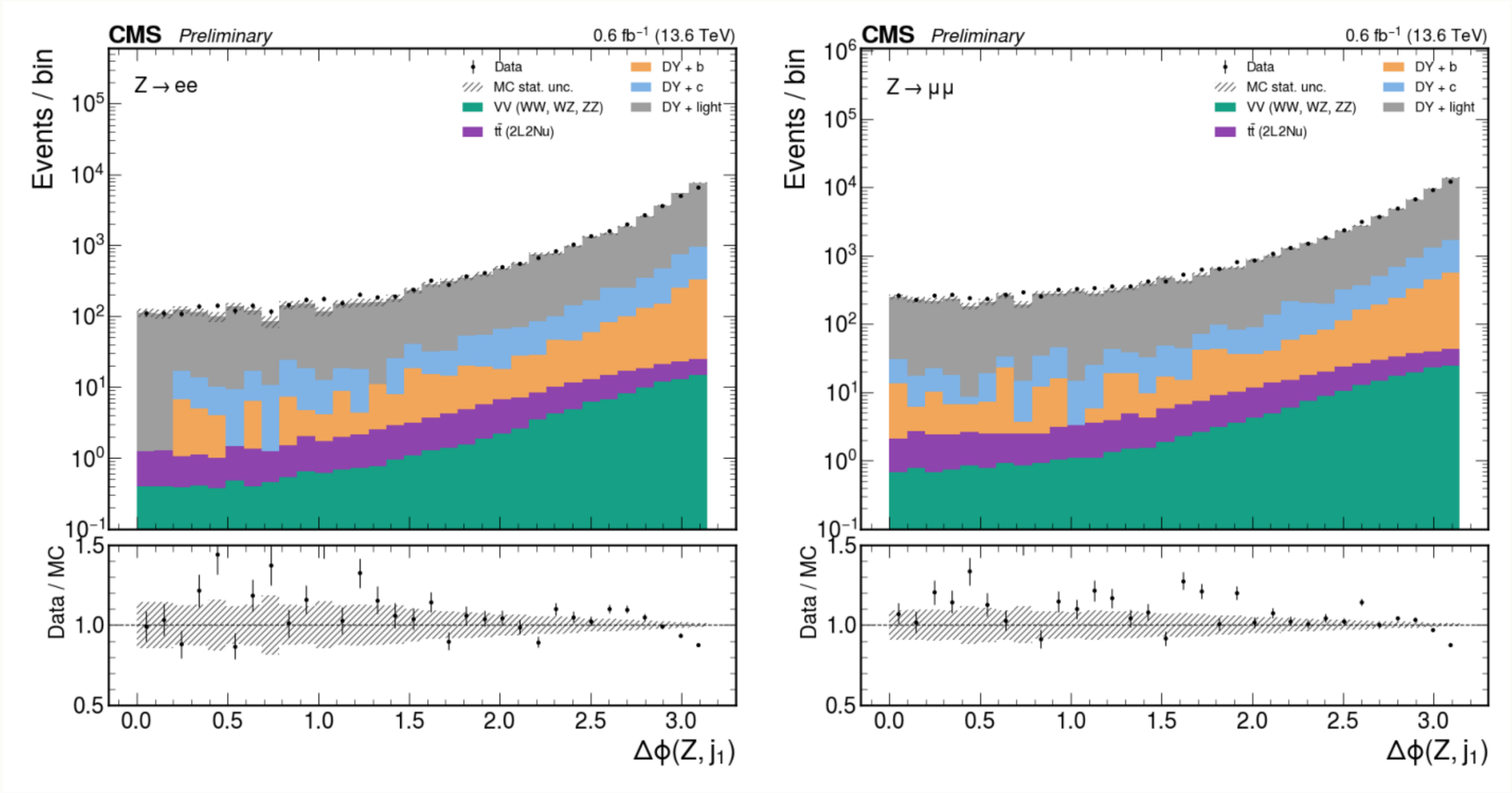
Events with at least two jets.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 34

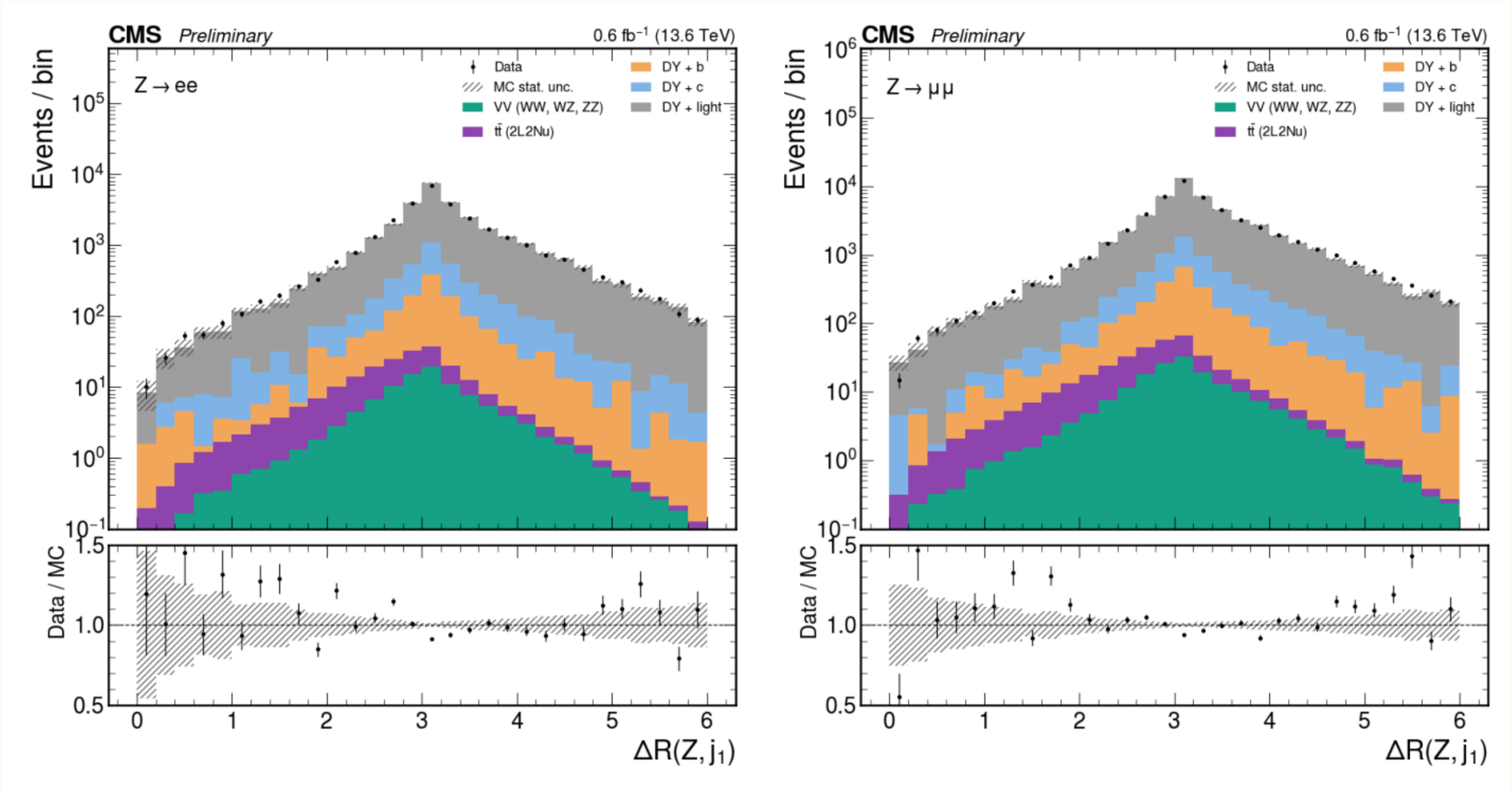
Δφ between the Z and the leading jet

Near π when the jet recoils against the Z.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 35

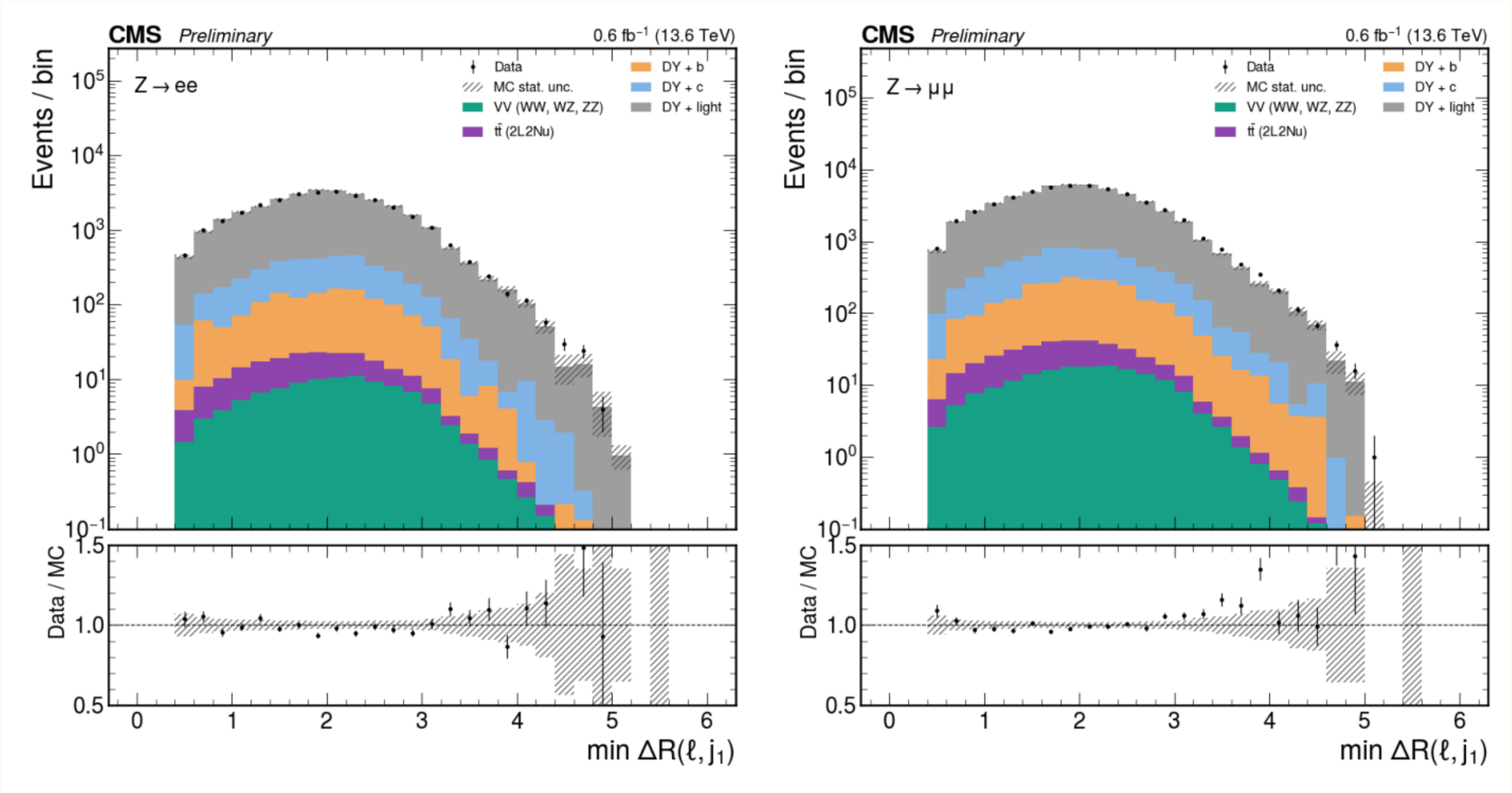
ΔR between the Z and the leading jet



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 36

Closest lepton to the leading jet

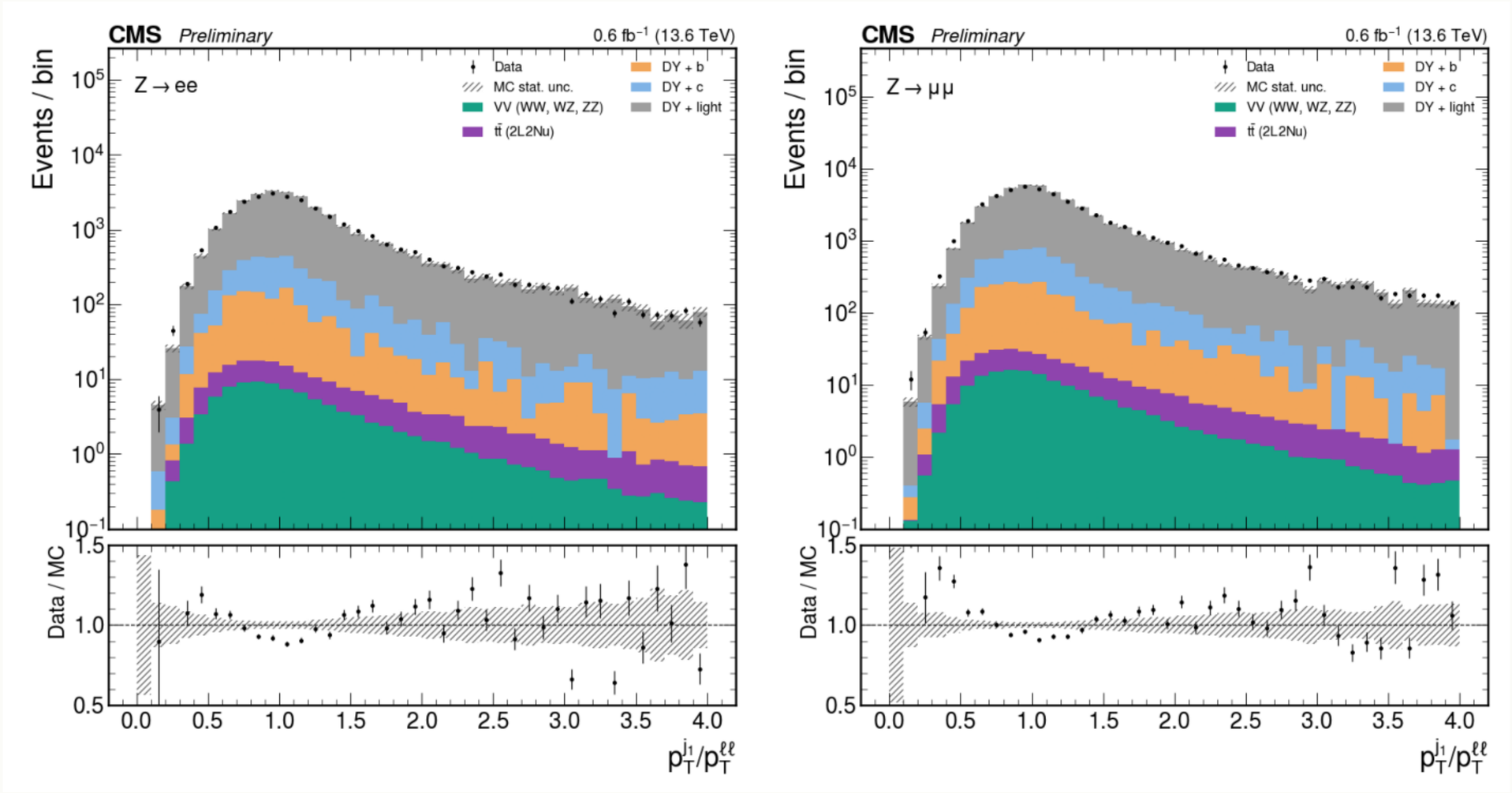
Smallest ΔR between the leading jet and a lepton. Jets within 0.4 of a lepton are removed.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 37

Jet pT / Z pT

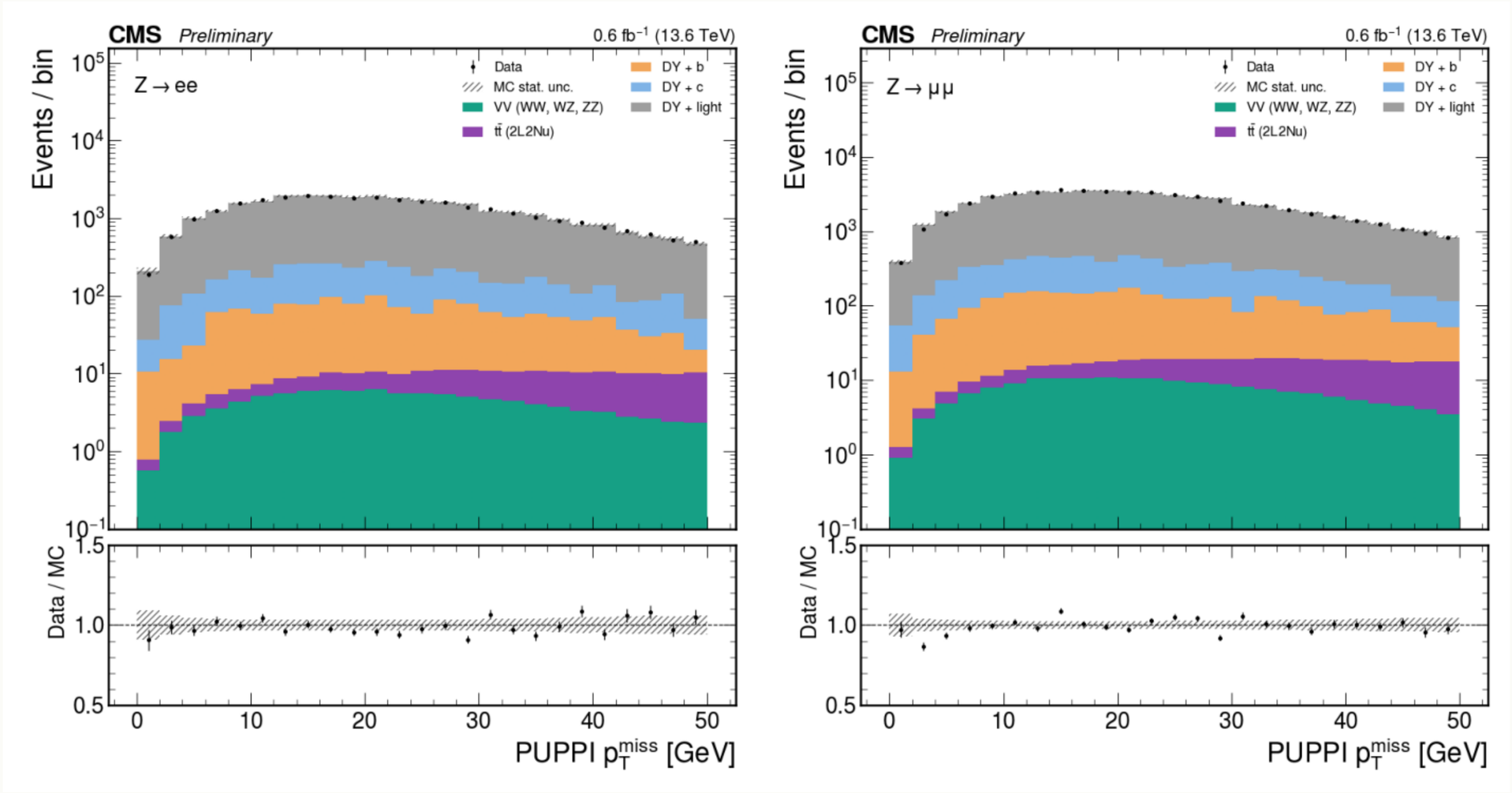
Leading-jet pT over Z pT. Near 1 when the jet balances the Z. Values above 4 are not shown.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 38

Missing transverse momentum

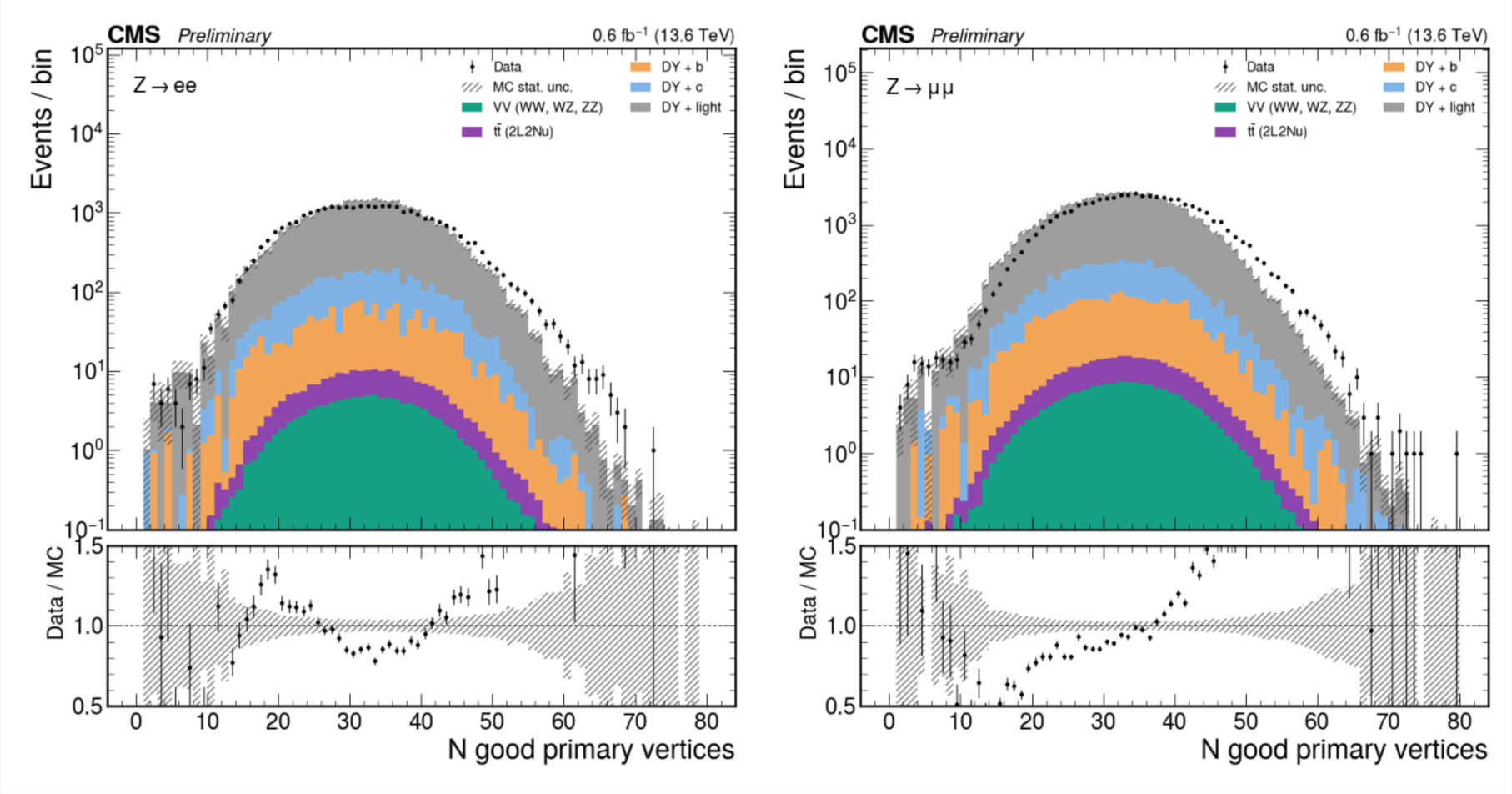
PUPPI MET, after the < 50 GeV cut.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 39

Number of primary vertices

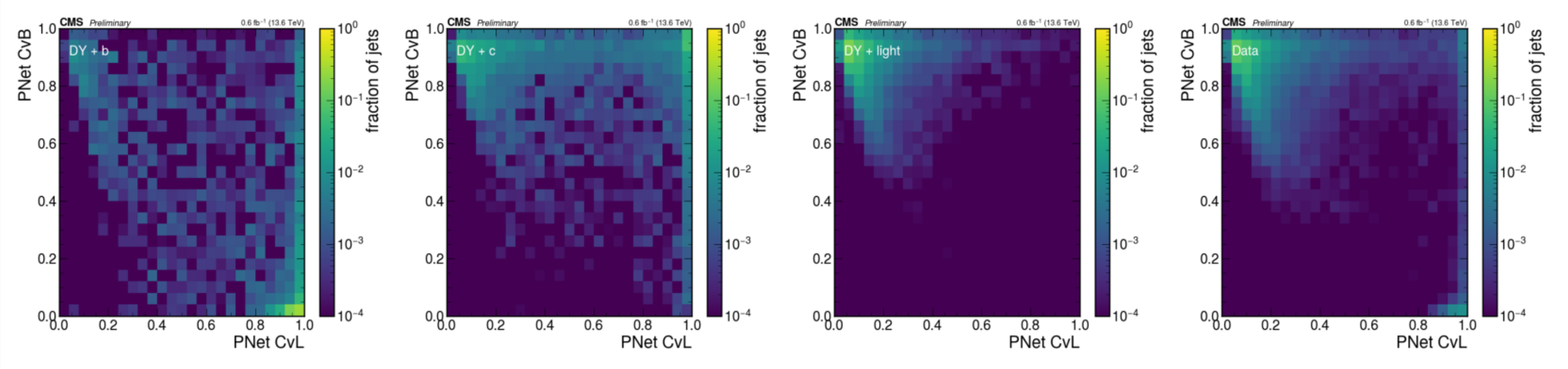
Good primary vertices. Each channel reads only 20 data files from a few runs, so its pileup is not that of the whole era. Most of the shape difference comes from that.



Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 40

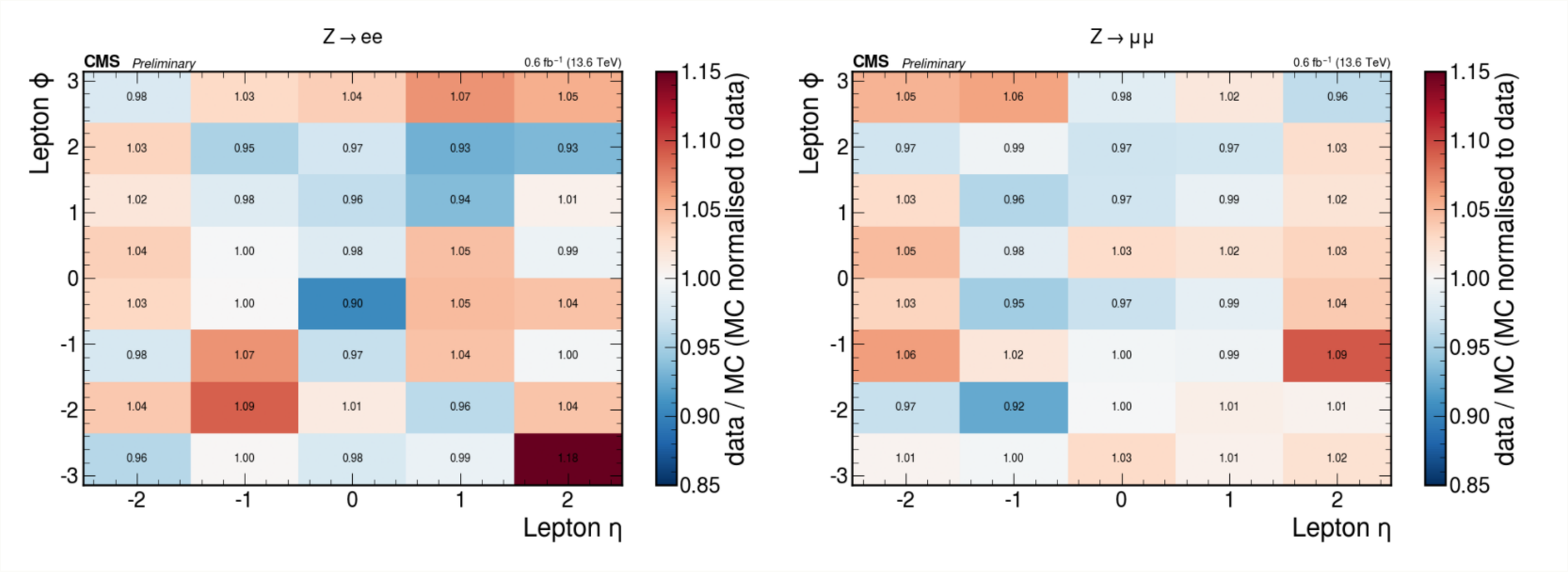
PNet charm-tagging plane: CvL vs CvB

Leading jets, both channels summed, each panel normalized to unit area. Data is a mix of all flavours, mostly light.



Where are the leptons? Data / MC in η and φ

Both Z leptons. MC normalised to data, so only the shape is compared. Cells are 1.0 in η by π/4 in φ. Blue: fewer events in data than in MC. Red: more.



Summary and next steps

Slope gone: data and MC agree within 1–2%

Two fixes in this run. The DY MC now gets the Z pT shape from Z→μμ data, and electrons no longer make fake MET (jets that are really an electron are left out of the MET, as in the JME recipe). Data/MC is now 0.99 ± 0.01 (Z→ee) and 1.00 ± 0.01 (Z→μμ), up from 0.97 and 0.99. Data/MC is flat in Z pT in both channels.

- Full run on condor, to get the pileup and jet-multiplicity shapes right.
- Muon trigger SF from both muons, like the electrons. Needs MC trigger efficiencies, which MUO does not provide.
- b/c-tag SFs once we cut on the tagger.